



Basic Principles of Medicine 1

Module: Cardiovascular, Respiratory and Renal

Lecture: CPR 28

Clinical Anatomy of the Respiratory System

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Recommended Reading



Drake R L, Vogl A W, Mitchell A W M. Gray's Anatomy for Students. 4th ed. Elsevier. Chapter 3, pps 166-183



Pay special attention to the green 'In the Clinic' boxes which contain the clinical correlates for this topic.

- Online Textbook: Chapter 3
 - Regional Anatomy
 - Pleural cavities
 - Pleura
 - Lungs

Netter Frank H. Atlas of Human Anatomy, 7th ed. Section 4, plates 202 -213

Objectives

SOM.MK.II.BPM1.3.CPR.2.ANAT.1439	Describe the locations of the parietal and visceral pleura and the areas where they meet.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1440	List the recesses formed by the pleural reflections.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1441	Differentiate between the thoracic cavity and the pleural cavity.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1442	Describe the relationship of the lungs to the pleural cavity.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1443	Describe and differentiate the surface projections (rib level) of the visceral pleura and those of the parietal pleura.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1444	List the identifying features of the right and left lungs and their surface projections.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1445	List the surfaces and borders of the lungs.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1446	Describe the surface projections (rib level) of lobes and fissures of the lungs.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1447	Identify the auscultation sites for lobes of the lung.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1448	Differentiate between the root and hilum of a lung.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1449	Describe the relationship of the structures in the root of each lung.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1450	Describe the typical branching pattern of the bronchial tree.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1451	Describe the arterial supply, venous and lymphatic drainage of the lungs and the bronchial tree.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1452	Describe a bronchopulmonary segment and its clinical importance
SOM.MK.II.BPM1.3.CPR.2.ANAT.1453	Describe the autonomic innervation of the lungs and bronchial tree including the specific response of each structure (glands, bronchial smooth muscle, and vascular smooth muscle) to the sympathetic and parasympathetic innervation respectively.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1454	Identify the major anatomical features of the lungs in imaging studies including but not limited to CT, MRI and radiographic images.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1455	Describe the embryological basis and clinical presentation of respiratory distress syndrome (hyaline membrane disease).
SOM.MK.II.BPM1.3.CPR.2.ANAT.1456	Compare and contrast the mechanism underlying a spontaneous vs traumatic pneumothorax.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1457	Compare and contrast the mechanism underlying an open pneumothorax vs a tension pneumothorax.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1458	Describe the clinical signs of a tension pneumothorax and explain why it is a medical emergency.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1459	Describe the decompression technique for a tension pneumothorax.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1460	Describe the location of pleural effusion versus pulmonary edema.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1461	Explain why an aspirated foreign body would usually pass into the right bronchus.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1462	Describe atelectasis.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1463	Describe the anatomical location, mechanism and clinical sequences of the insertion of an endotracheal tube.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1464	Describe a Pancoast tumor and explain how it can lead to thoracic outlet syndrome.

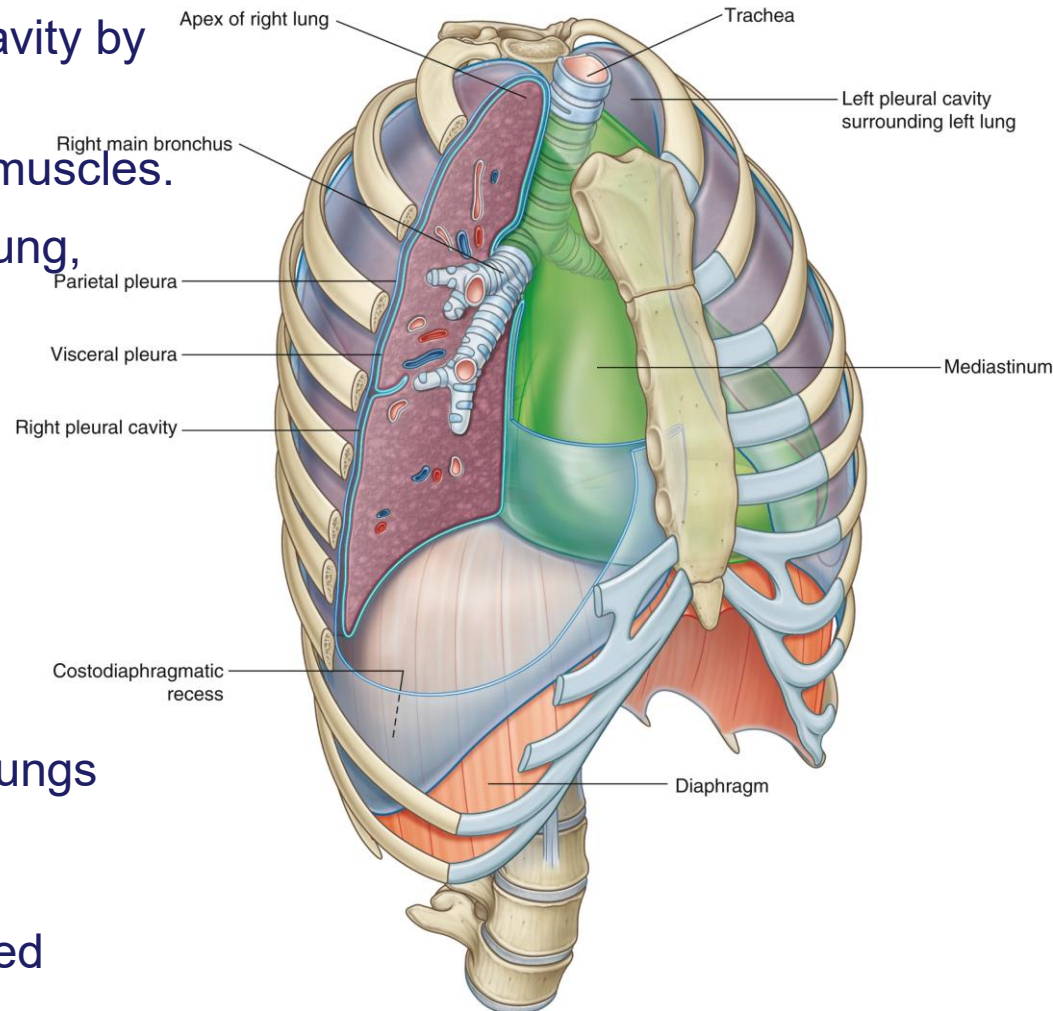
Thoracic and Pleural Cavities

Thoracic cavity

- Space found in the upper trunk region.
- Separated from the abdominal cavity by the diaphragm.
- Protected externally by ribs and muscles.
- House important organs (heart, lung, esophagus)
- Contains the mediastinum and three serous cavities
- two pleural cavities
- one pericardial cavity

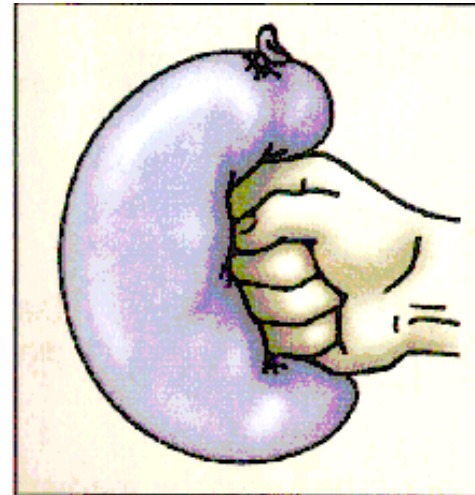
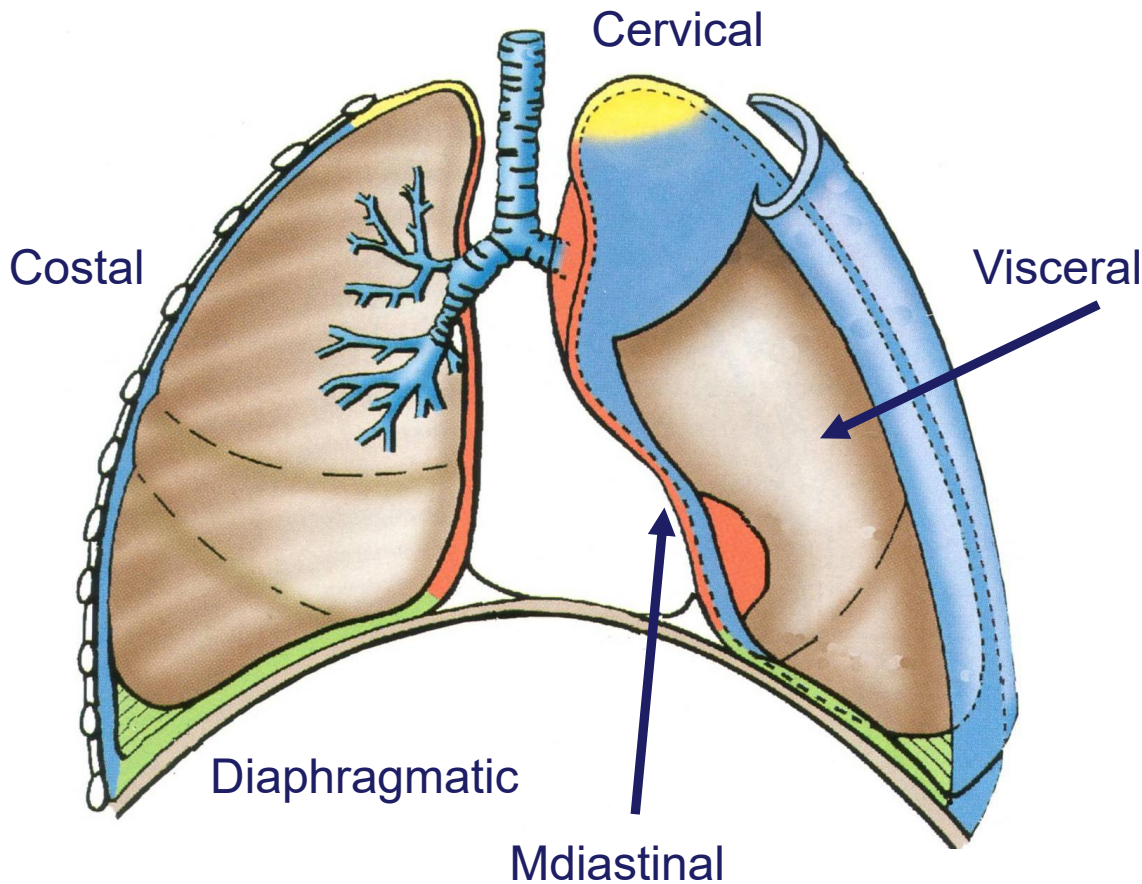
Pleural cavity

- Closed cavities surrounding the lungs situated on either side of the mediastinum
- Lined by serous membranes called pleura

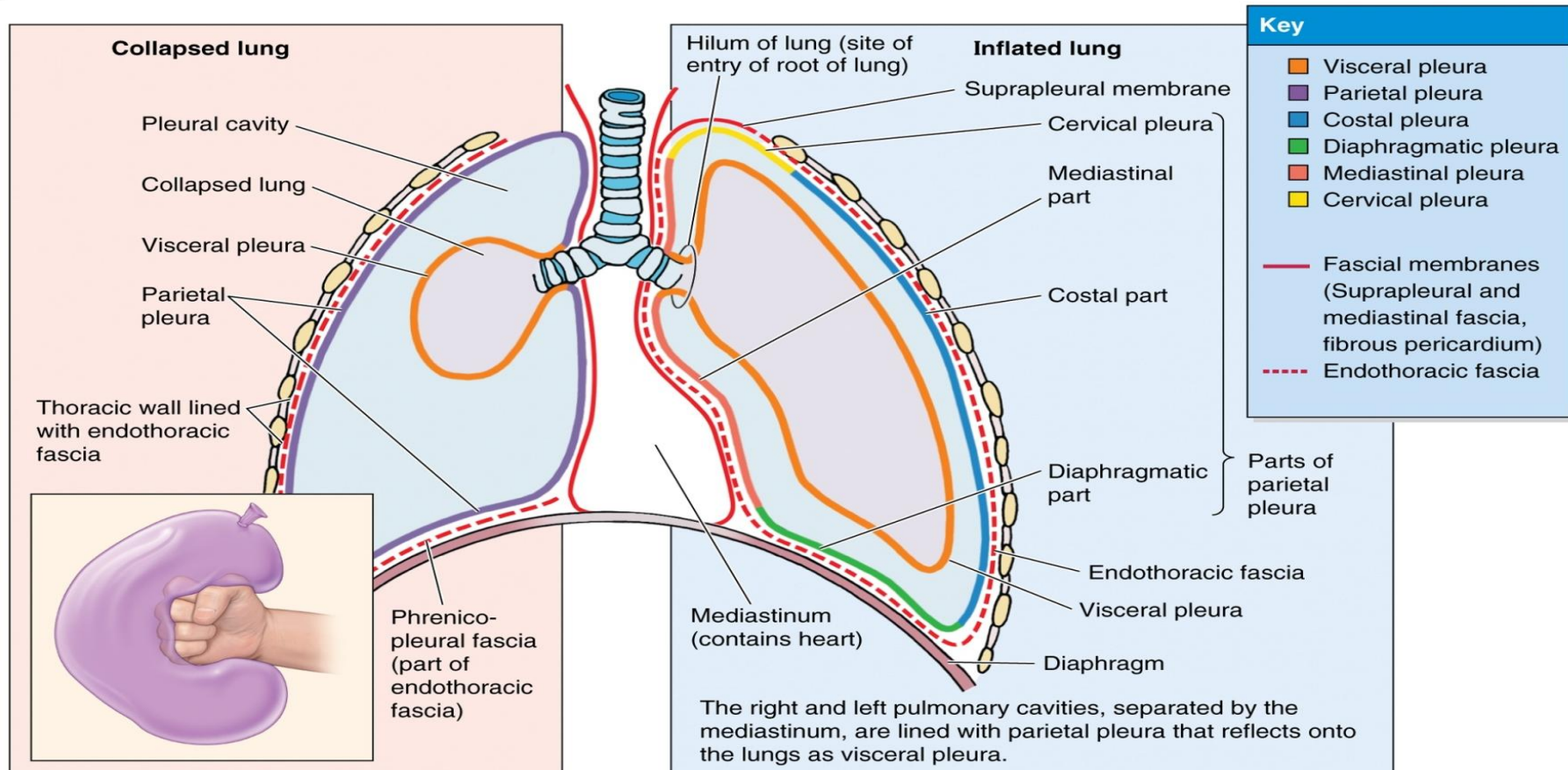


Pleura

- is a serous membrane which surrounds each lung (Bursa like fluid bags) and is a continuous closed system
- has two layers
 - visceral pleura – invests the lungs (adherent to organ)
 - parietal pleura – covers the internal surface of thoracic wall (away from organ)



Parietal and Visceral Pleura



Pleura Cavity

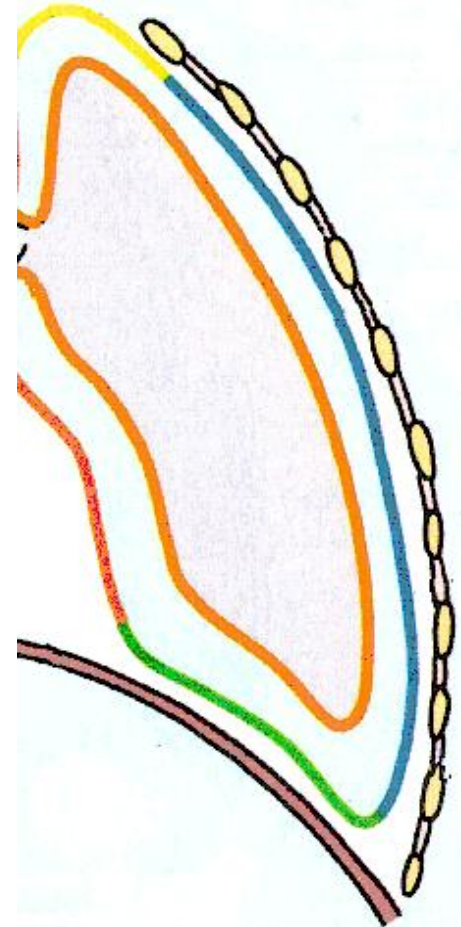
- Potential space between visceral and parietal pleura
- Normally contains only a thin film of serous fluid

Pleural recess – is a potential space where two adjacent parietal pleura are in contact (during expiration) and fill with lungs during deep forced inspiration.

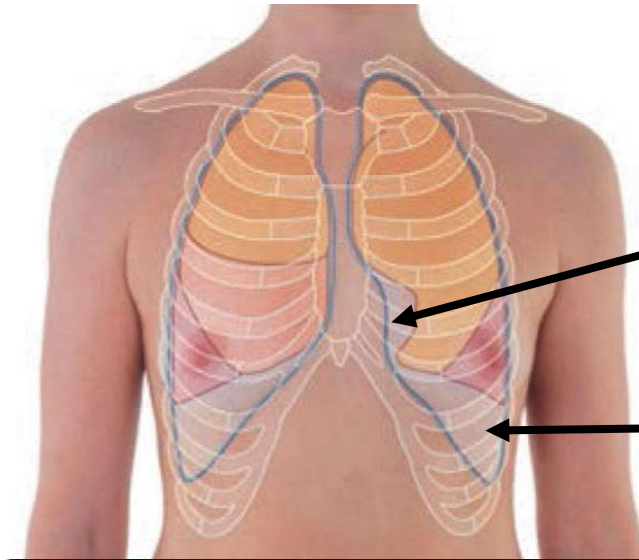
The recesses also provide potential spaces in which fluids can abnormally collect and from which fluids can aspirated

Pleural recess are :

- **Costodiaphragmatic**
- **Costomediastinal**



Pleural Recesses

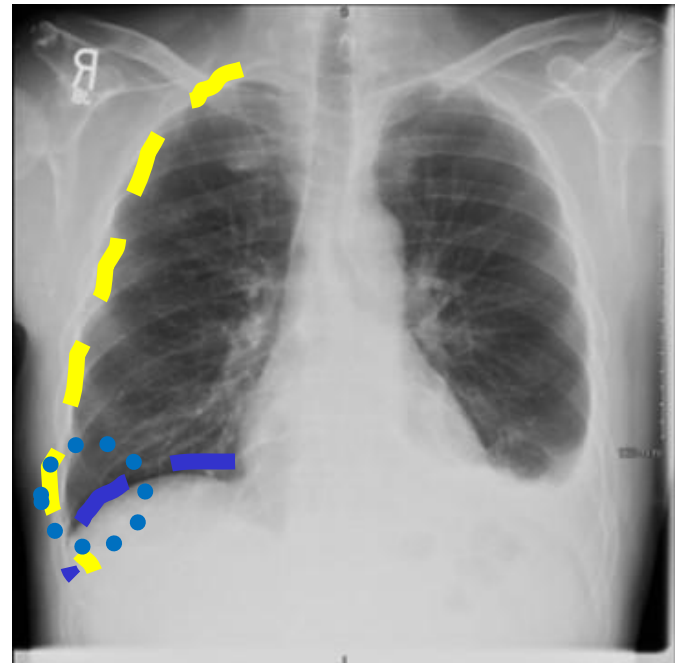


Costomediastinal recess

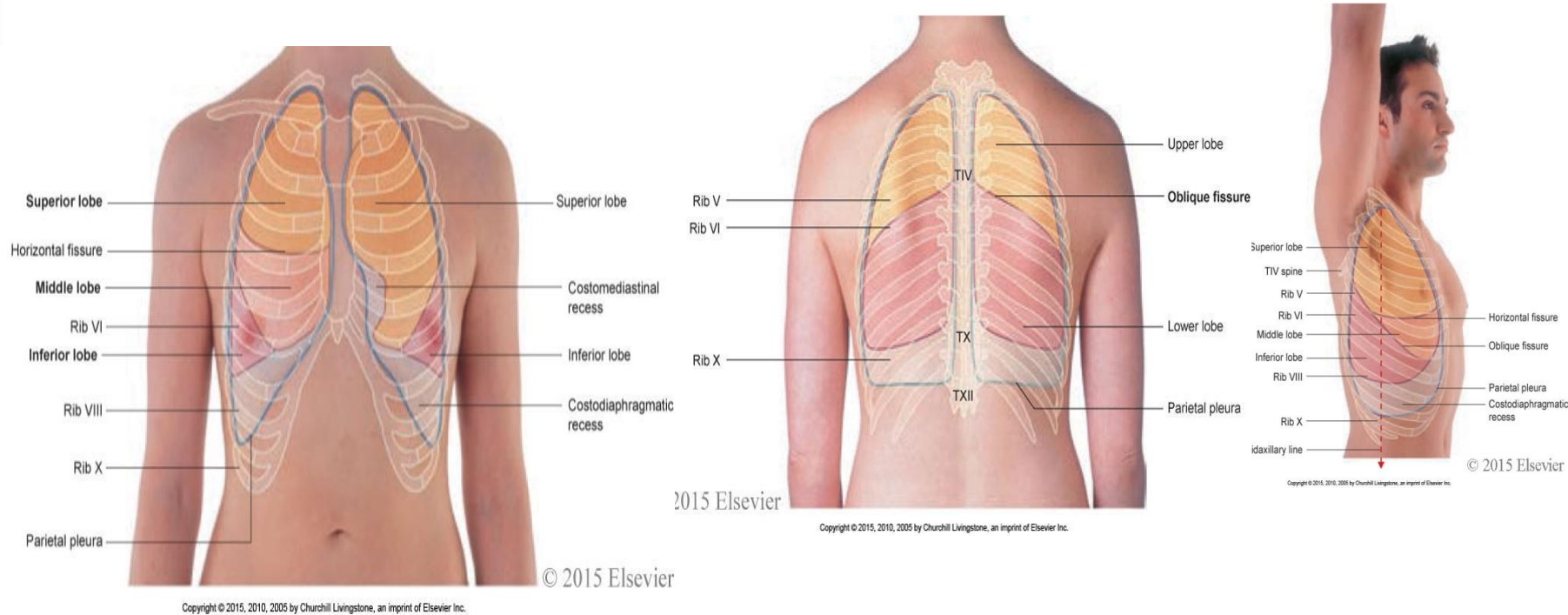
between costal and mediastinal
pleura

Costodiaphragmatic recess

between costal and diaphragmatic
pleura



Surface Landmarks of Pleura



*At rest (exhaled)	Inferior border of lung	Inferior edge of parietal pleura
Mid-clavicular line	Rib 6	Rib 8
Mid-axillary line	Rib 8	Rib 10
Paravertebral	TV10	TV12

Between these points is the costodiaphragmatic recess (ideal site for accessing pleural cavity - thoracentesis etc.)

Innervation of the Pleura

The parietal pleura is supplied by the:

- **intercostal nerves** (costal and cervical pleura),
- **phrenic nerves** (mediastinal and central diaphragmatic pleura)
- **lower five or six intercostal nerves** (peripheral diaphragmatic pleura).

The visceral pleura is insensitive to pain and touch due to its autonomic innervation

Pneumothorax

Pneumothorax- presence of air in the pleural cavity

Cause :

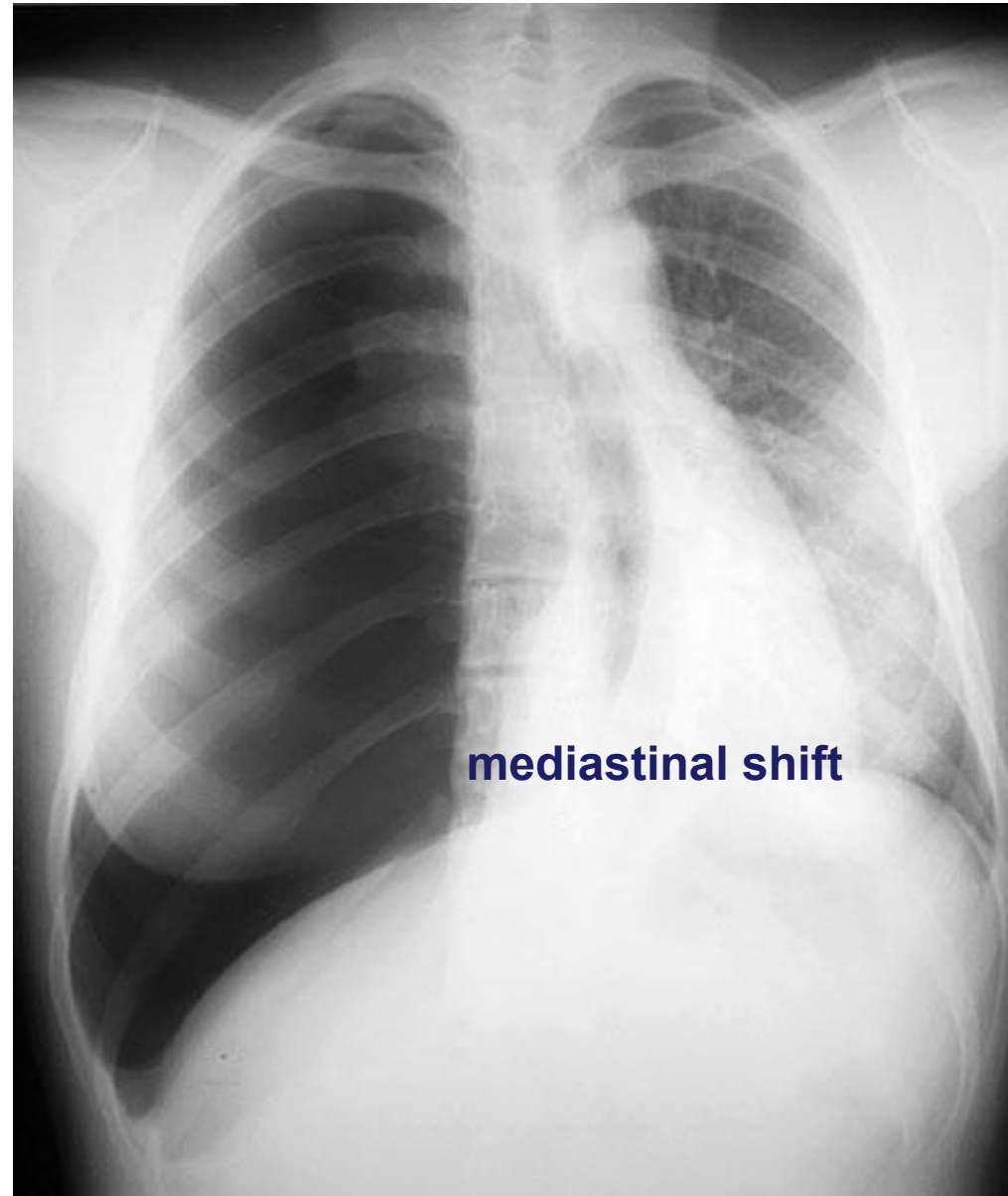
Spontaneous-

Absence of lung disease, no prior provoking event, ruptured bleb or bullae

Traumatic- introduction of air in the pleural cavity secondary injury to the pleura by blunt or penetrating trauma

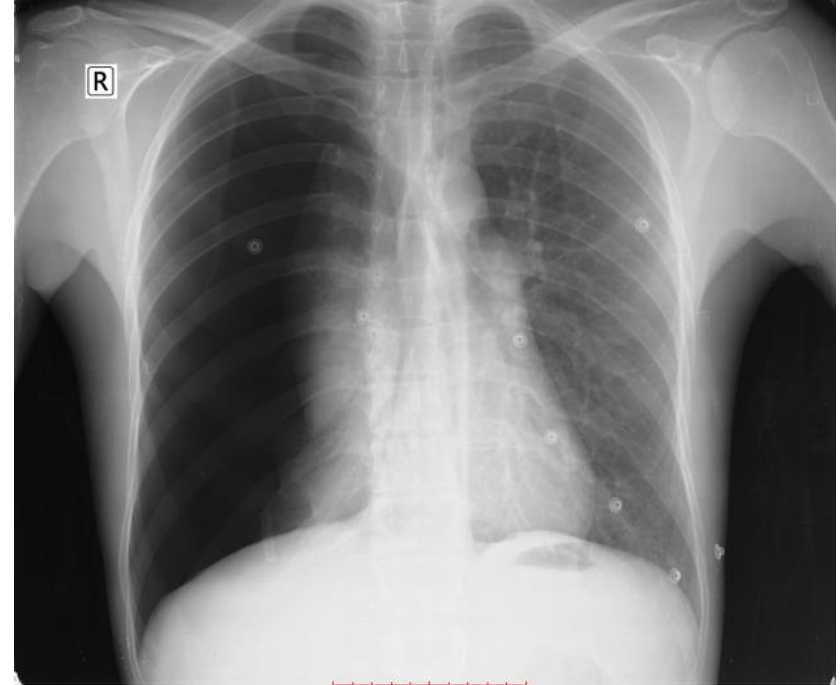
Other causes

- Inflammation
- Smoking
- Underlying pulmonary disease.



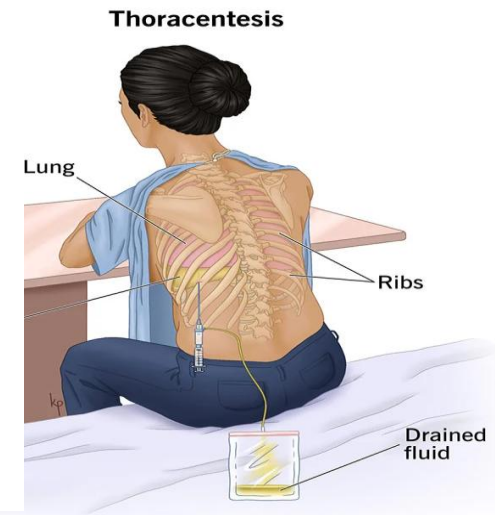
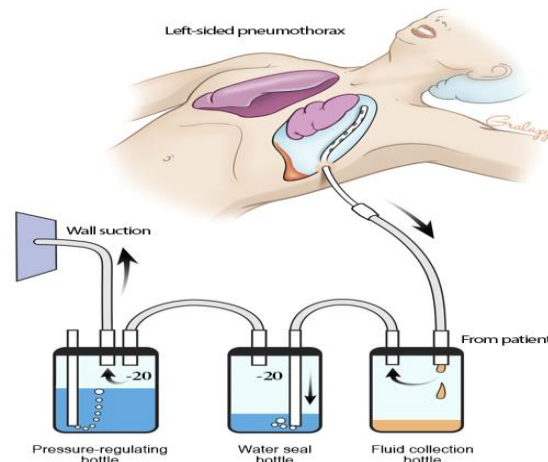
Pneumothorax

- **Non tension pneumothorax** there is no valve mechanism (unsealed opening) as a result there is no build up pressure.
- **Tension pneumothorax** is the condition in which the air filling the pleural cavity can not escape (forming one-way valve). In this condition the visceral pleura are ruptured. The pressure in the pleural cavity builds up with every breath causing **mediastinal shift**. This condition leads to severe shortness of breath, as well as circulatory collapse and requires urgent intervention.



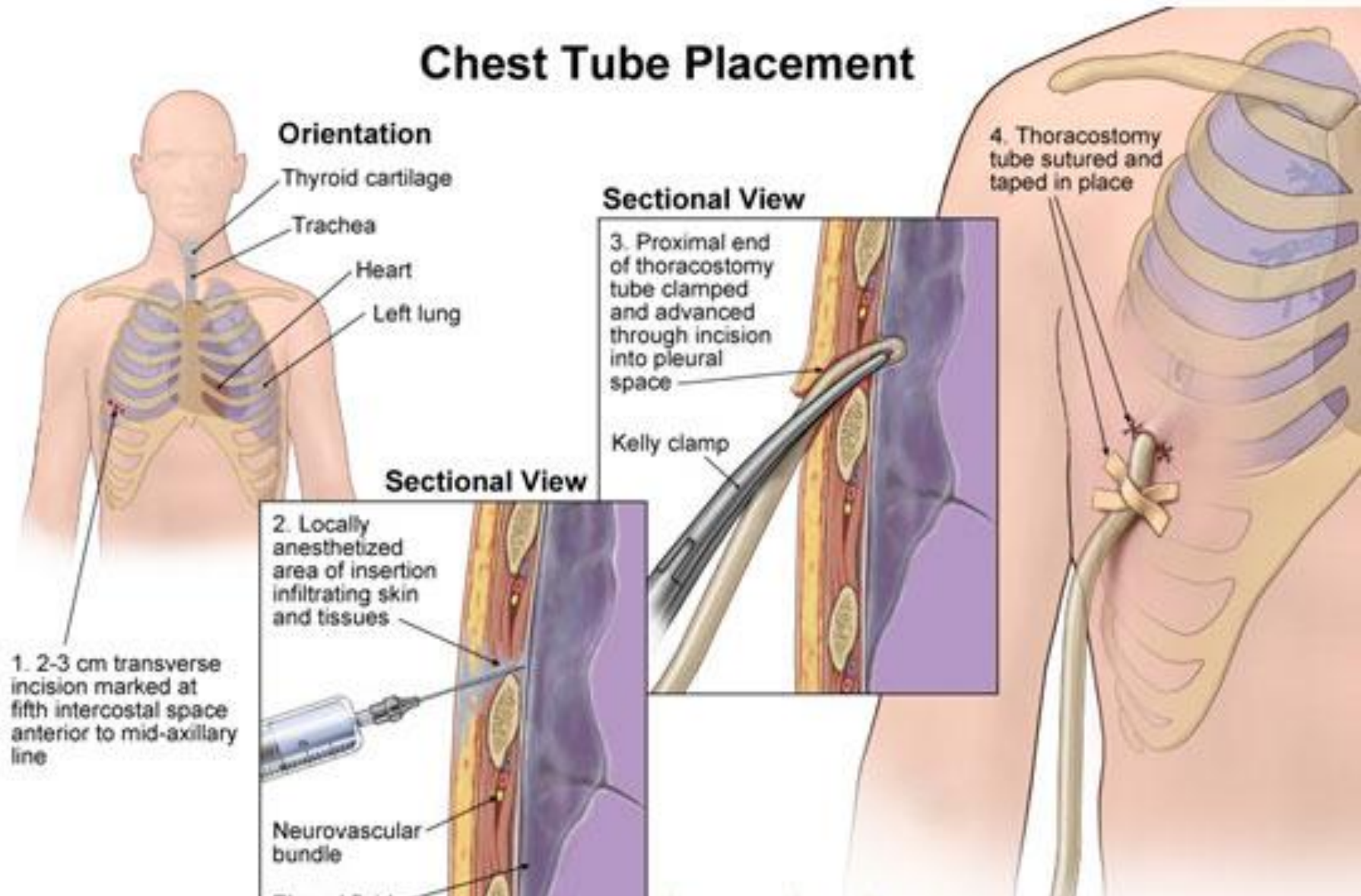
Decompression of Tension Pneumothorax

	Needle decompression	Tube Thoracostomy (Chest tube)
Indication	Tension pneumothorax -when thoracostomy is not possible in prehospital setting. Tube thoracostomy should follow.	relieve trapped collections of air or fluid in the thorax
Site	2 nd intercostal space in midclavicular line in affected hemithorax.	4 th or 5 th intercostal space between the anterior axillary and midaxillary
Equipment	Large bore needle- 14/16 gauge	Chest tube



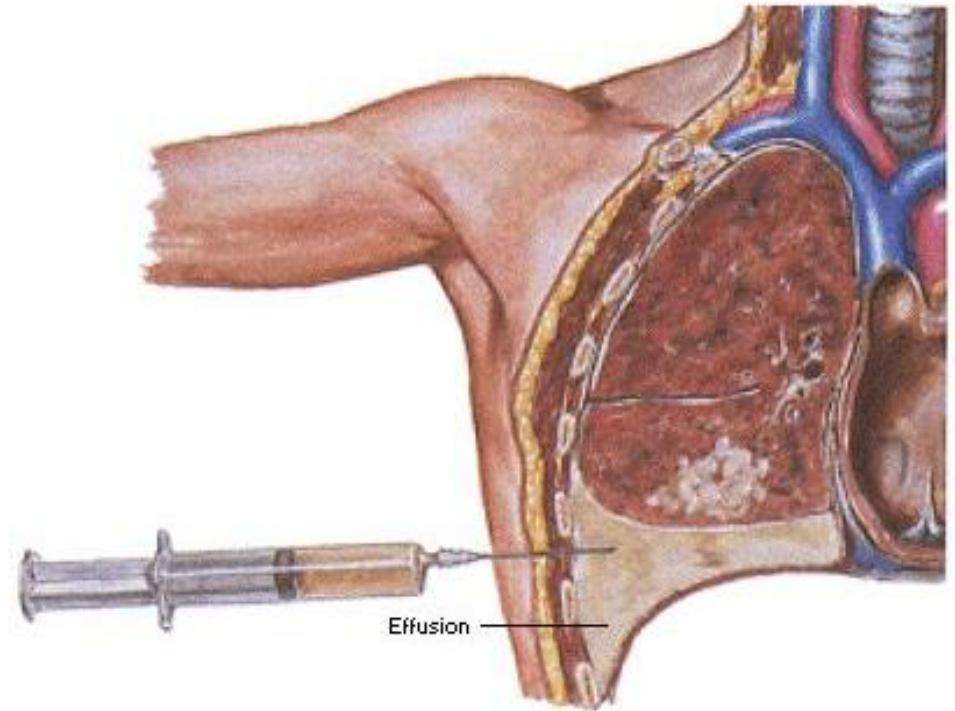
Thoracostomy

Chest Tube Placement



Thoracocentesis

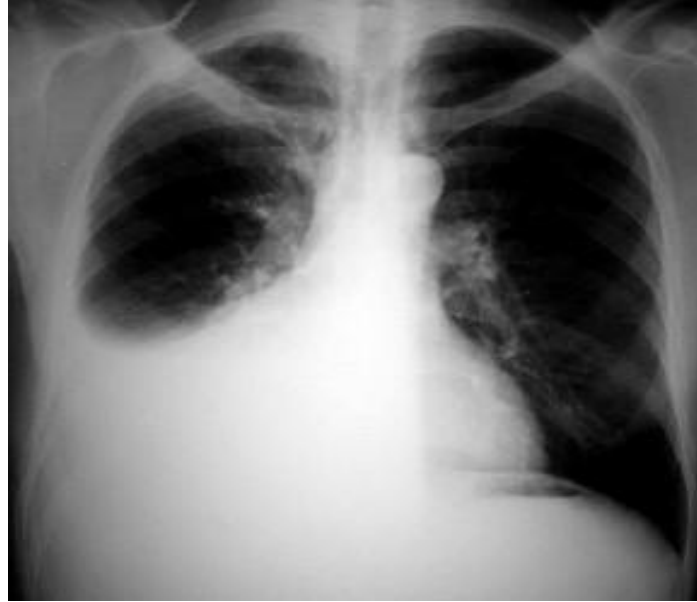
- Remove excess fluid or to take a sample of fluid.
- Insert the needle along the superior border of the rib (VAN – inferior edge of rib)
- Depending the amount of fluid and the distribution into the pleural cavity



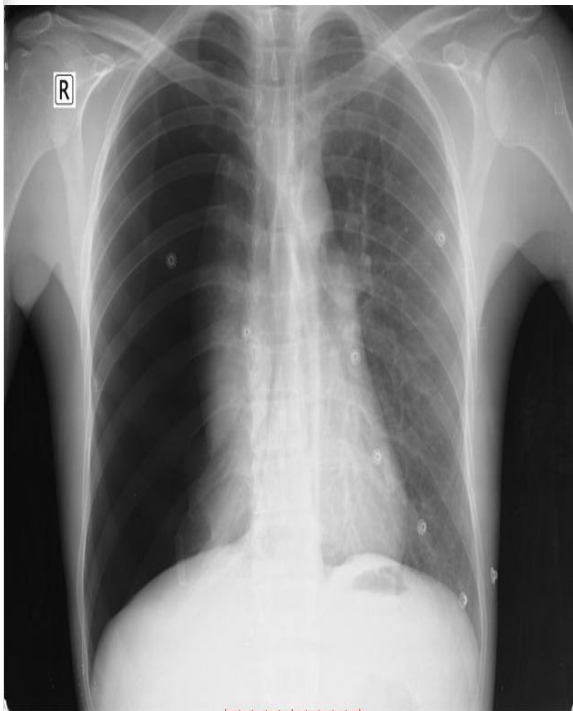
Thoracocentesis performed to obtain sample for testing

Pleural Fluid

- **Hemothorax** is the accumulation of **blood** within the pleural cavity
- Result of trauma such as a **stab wound** or **fractured rib** but can also be due to lung disease.
- **Pleural effusion** is the accumulation of inflammatory exudate within the pleura cavity
- The pain is referred from the parietal pleura to the cutaneous distribution of the intercostal nerve.



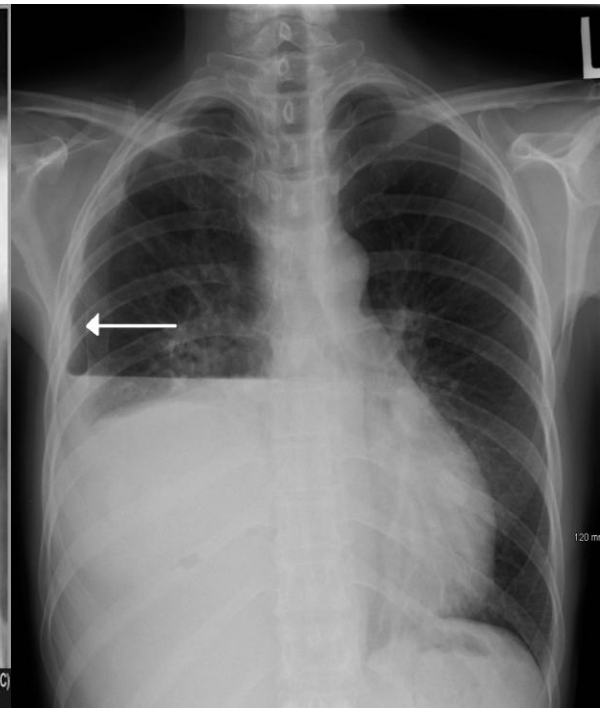
Pneumothorax



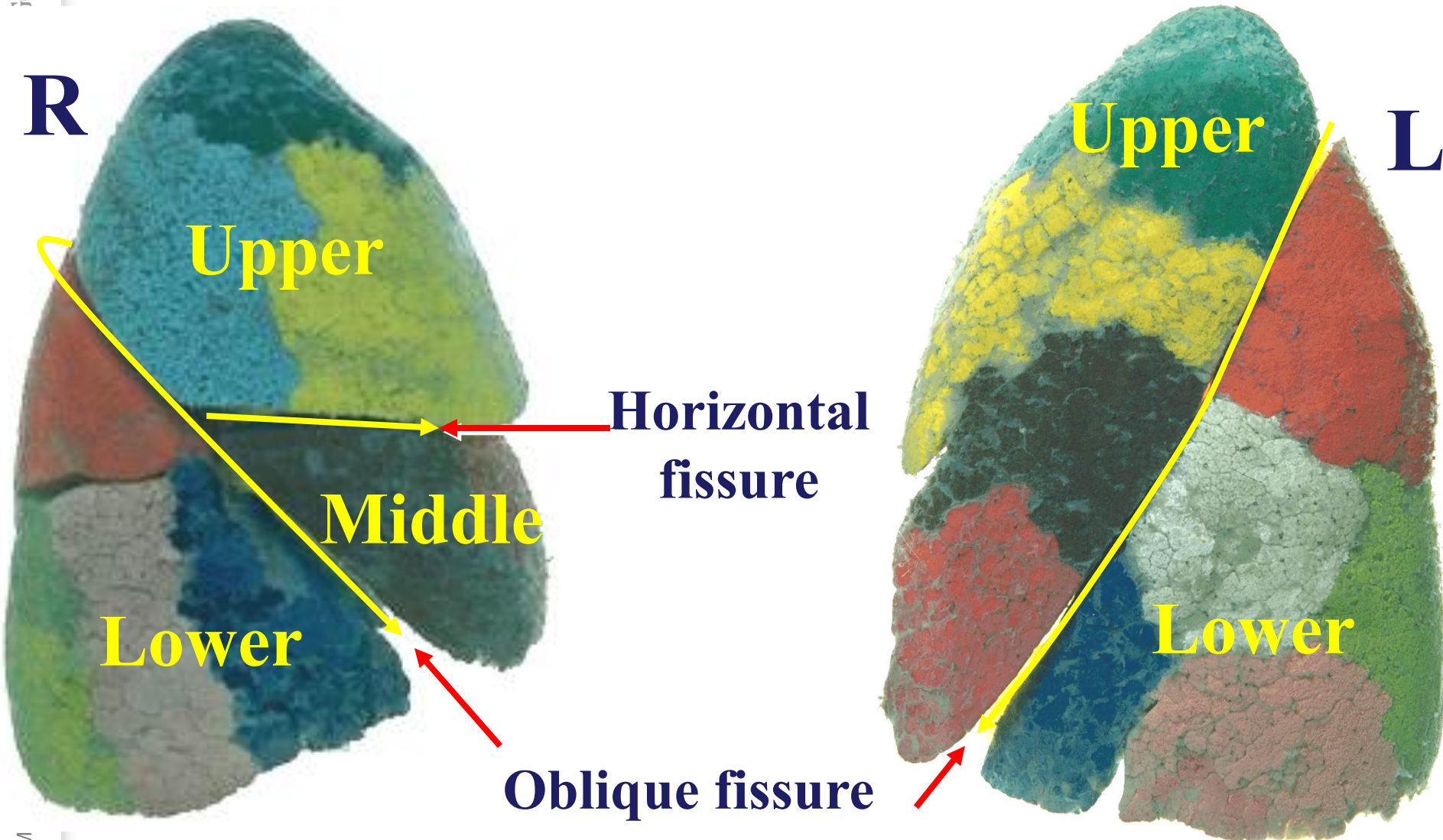
Pleural effusion



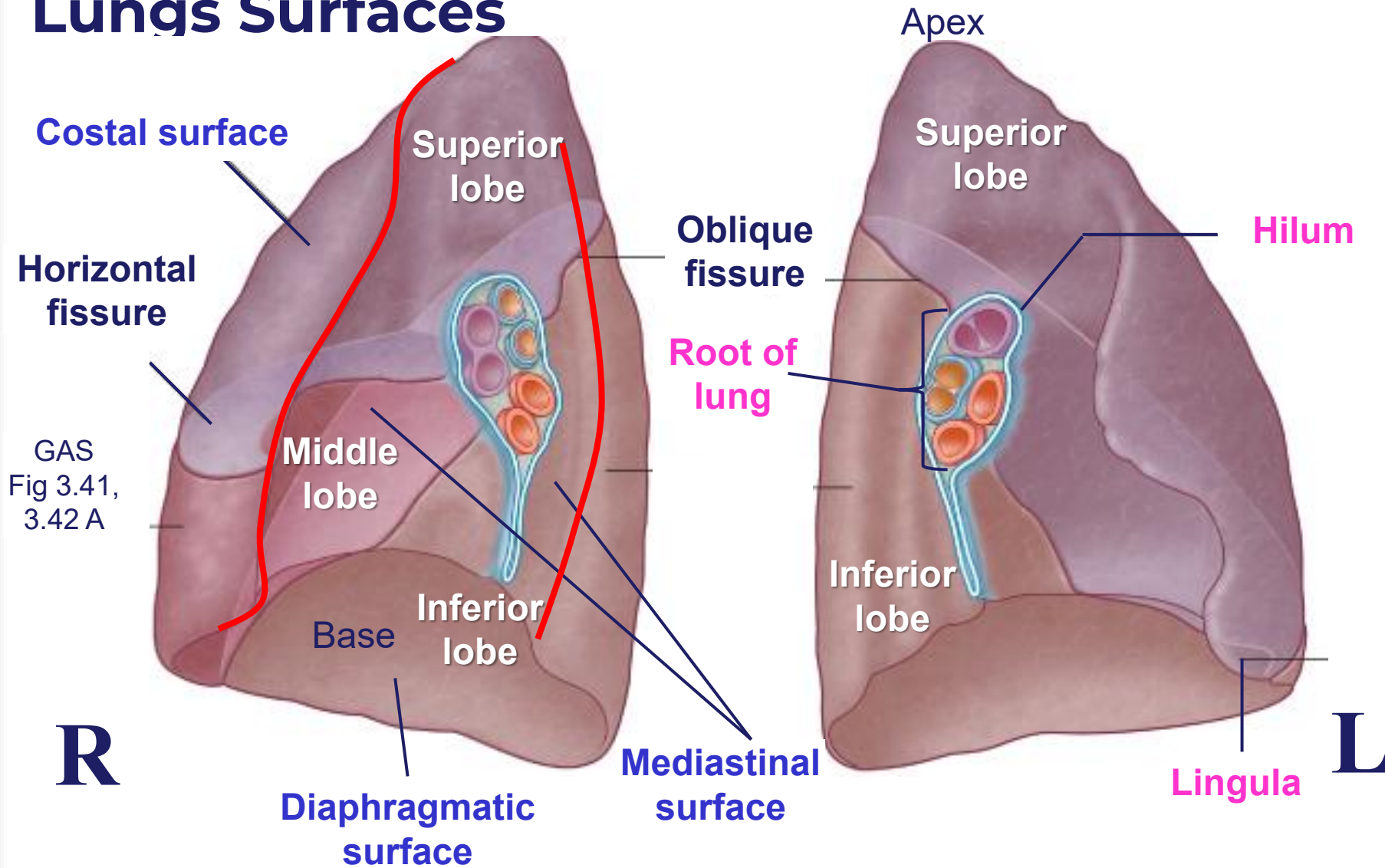
Hydropneumothorax



Lung Lobes and Fissures



Lungs Surfaces



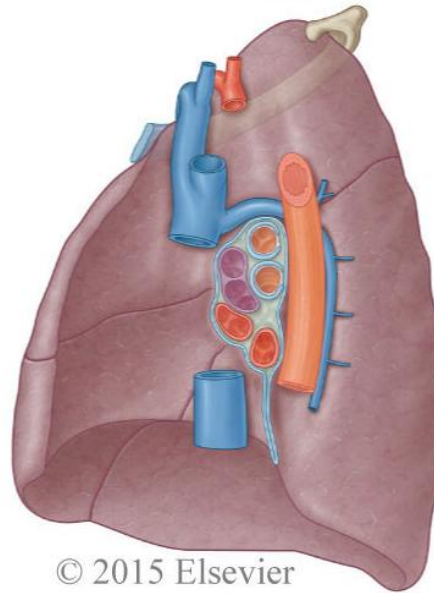
Root: Attachment of the lung to the structures in the mediastinum.

Hilum: the mediastinal pleura meets the visceral pleura.

Lung Relationships

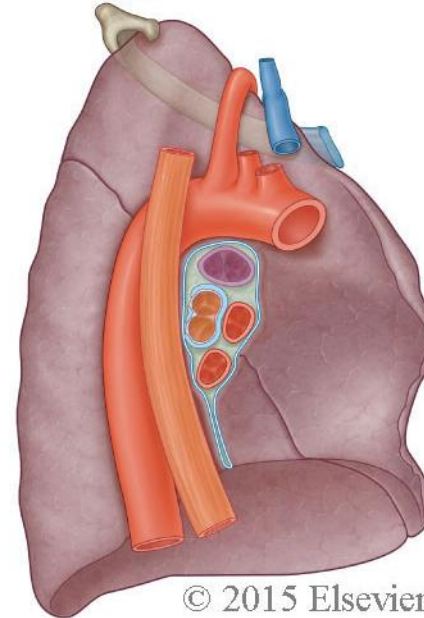
Medial Surface (Right Lung)

- SVC
- Heart
- IVC
- Azygos v
- Esophagus



Medial Surface (Left Lung)

- Heart
- Aortic arch
- Thoracic aorta
- Esophagus



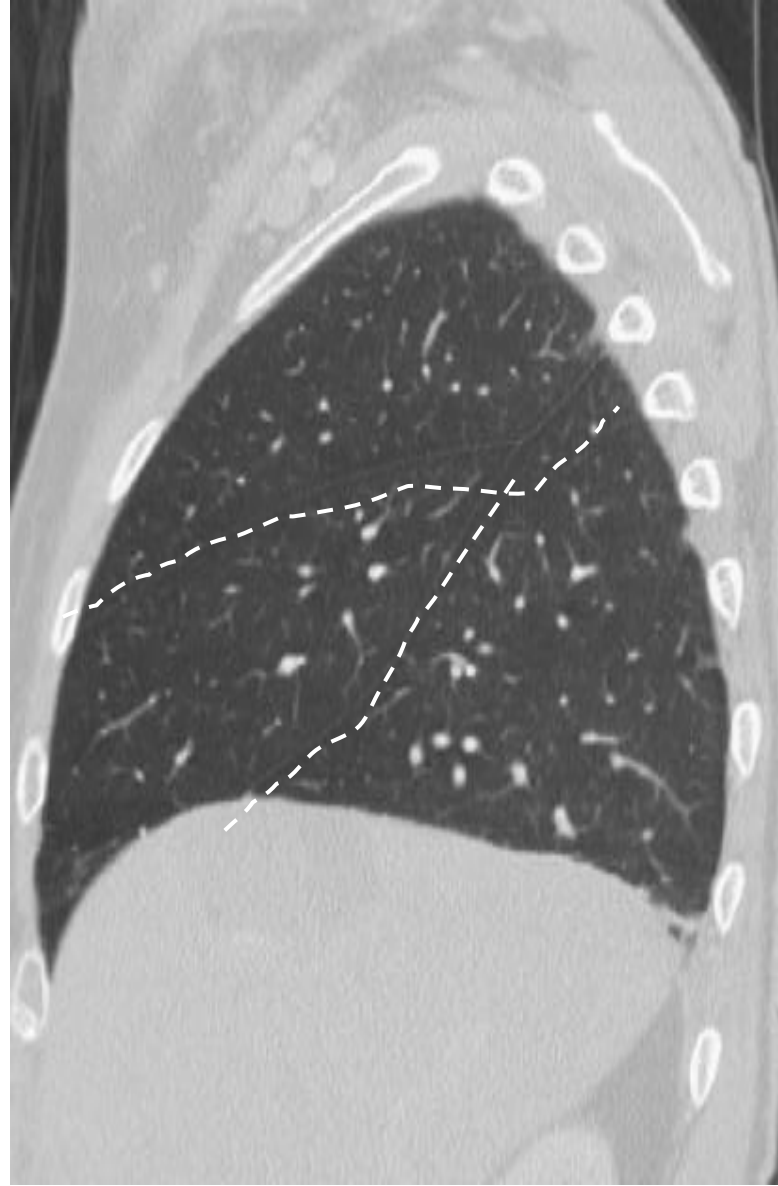
Hilum

- Region where the mediastinal pleura meets the visceral pleura.

Root

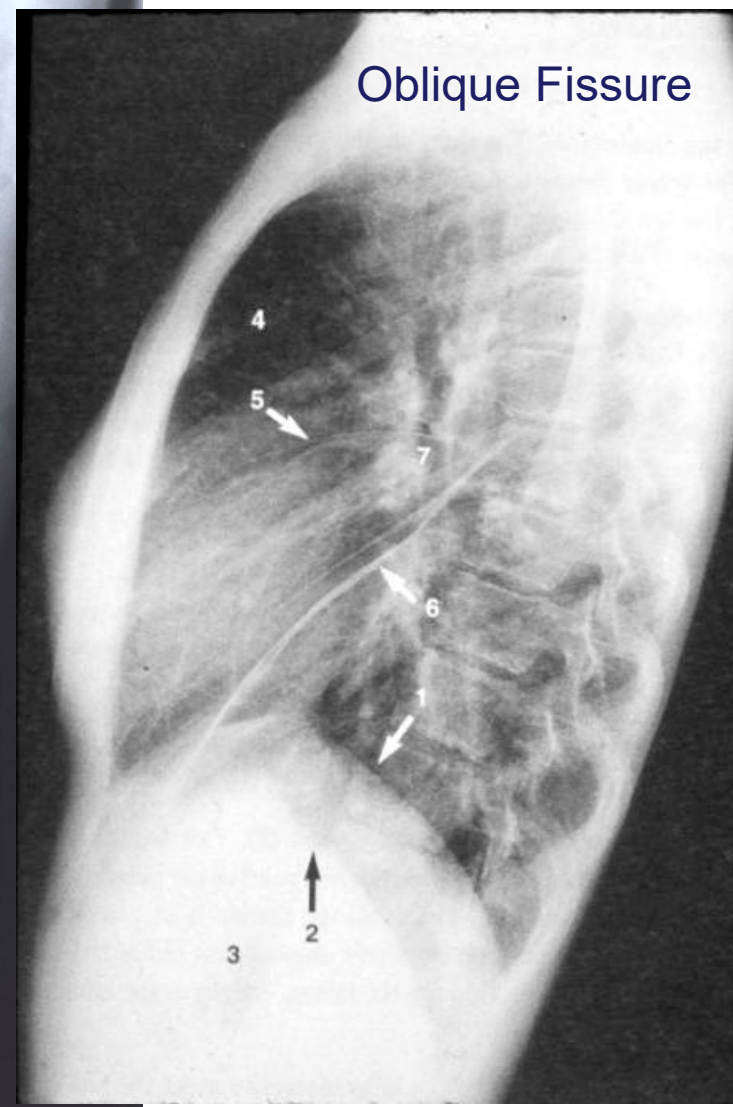
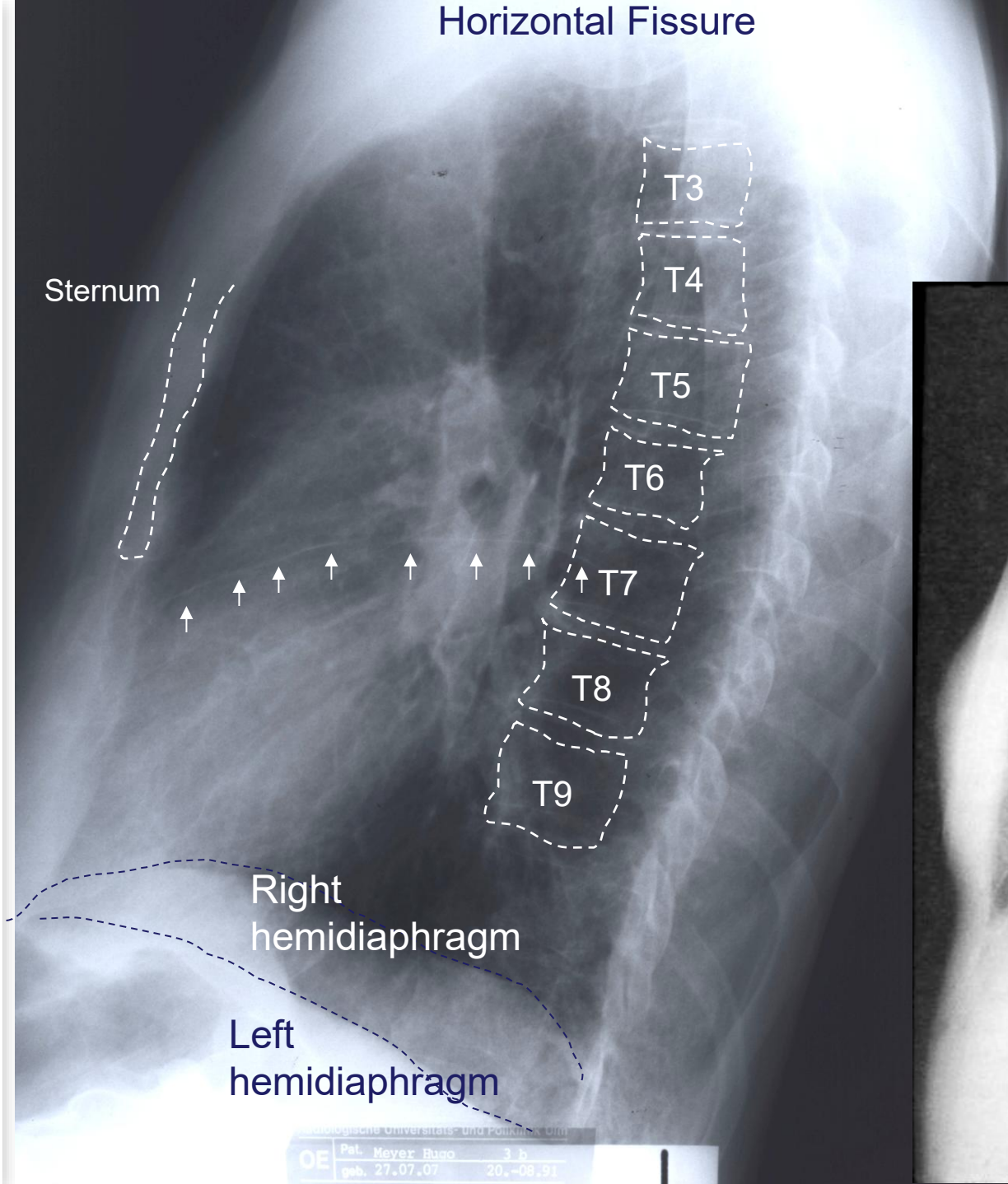
- Attaches the lung to the structures in the mediastinum.
- Formed by structures entering and leaving the lung such as:
 - Bronchi, Pulmonary artery(venous blood), Pulmonary vein(arterial blood), Bronchial arteries, Lymph vessels , Nerves

Lung Fissures



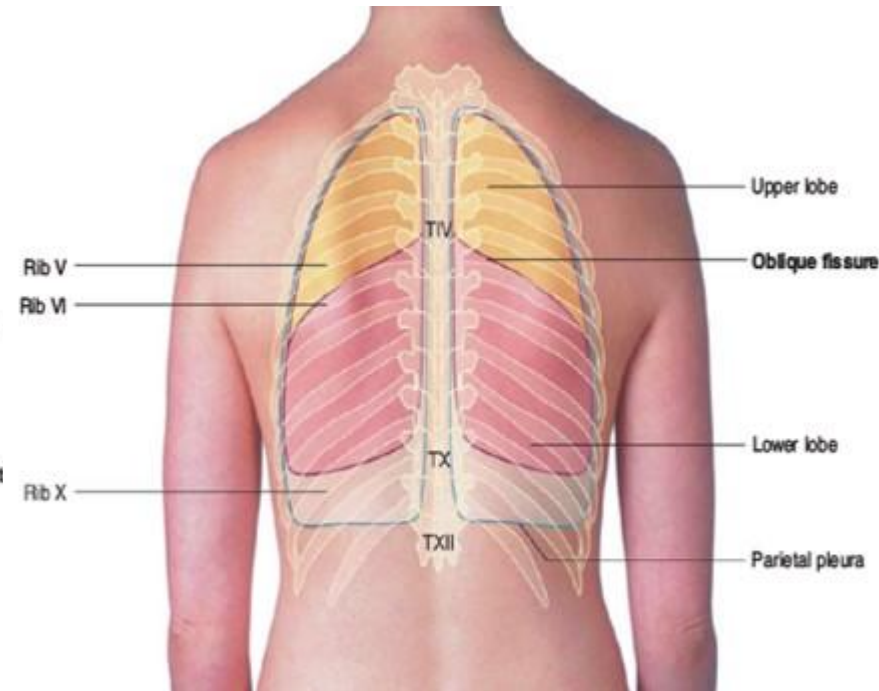
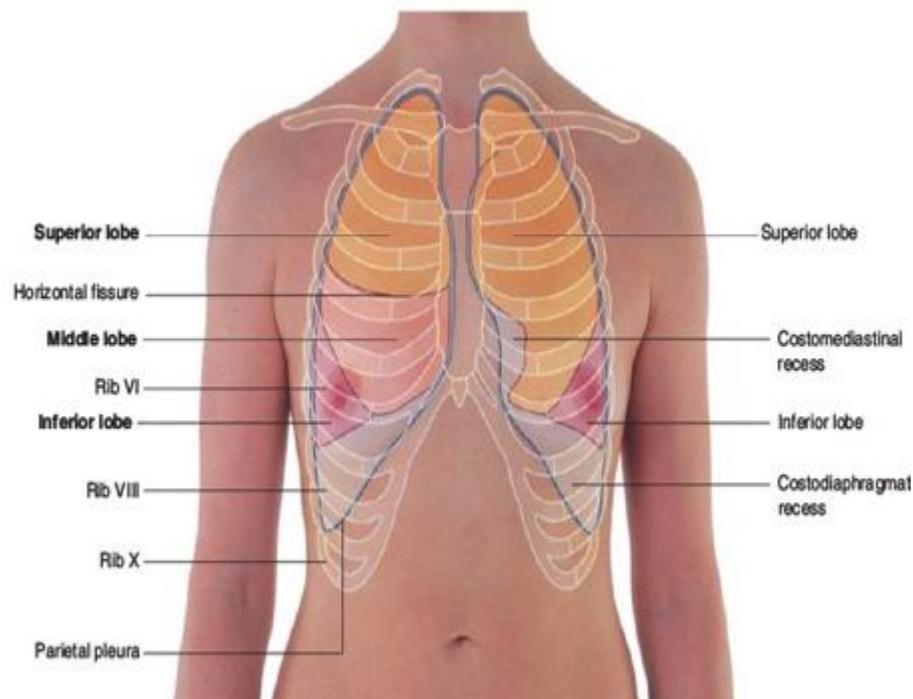
Horizontal Fissure

SOM.MK.II.BPM1.3.CPR.2.ANAT.1454



Surface Projections

- Oblique fissures: begin at the spinous process of the scapula (T4) and follows rib T6 anteriorly.
- Horizontal fissure: follows T4 intercostal space laterally to meet oblique fissure.



Bronchial Tree

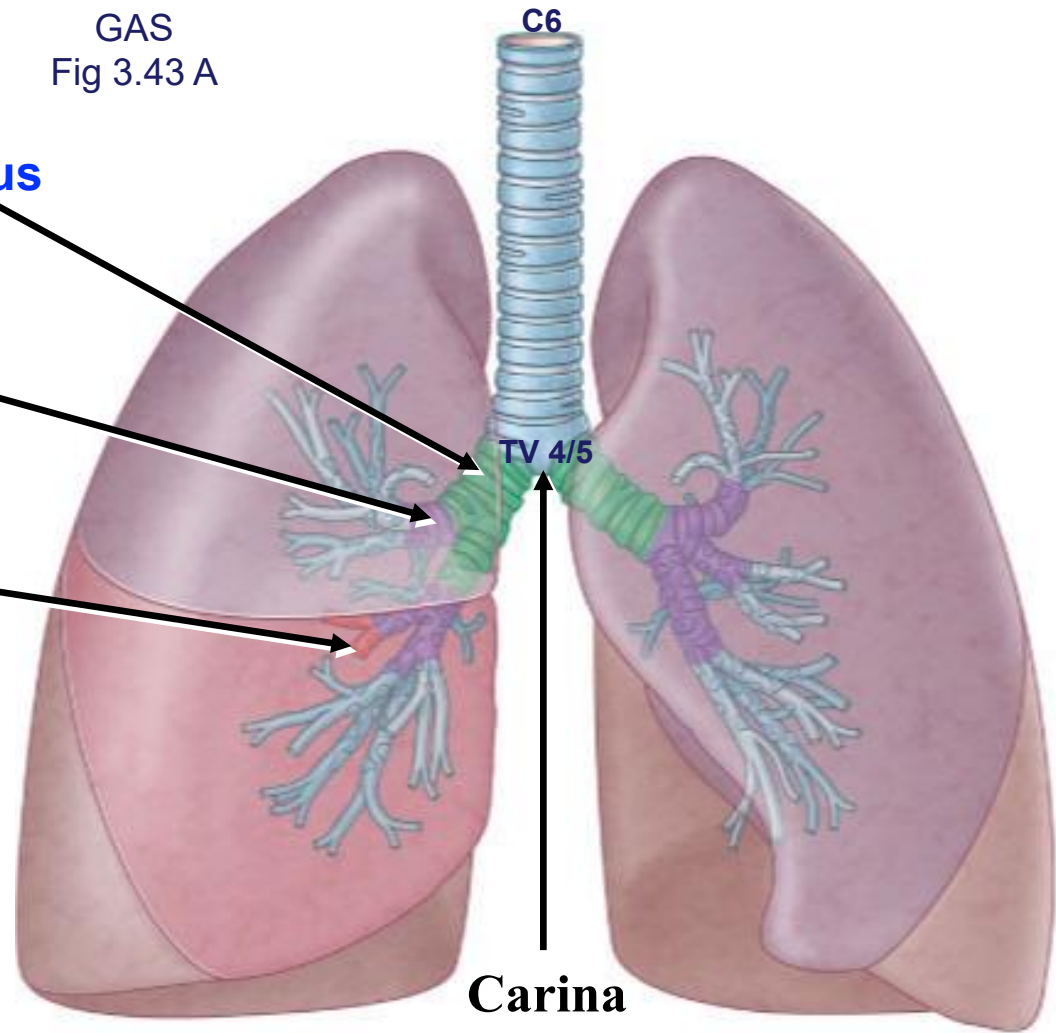
GAS
Fig 3.43 A

Primary (main) bronchus
- To lung

Secondary (lobar) bronchus
- To lobe

Tertiary (segmental) bronchus
-To bronchopulmonary segment

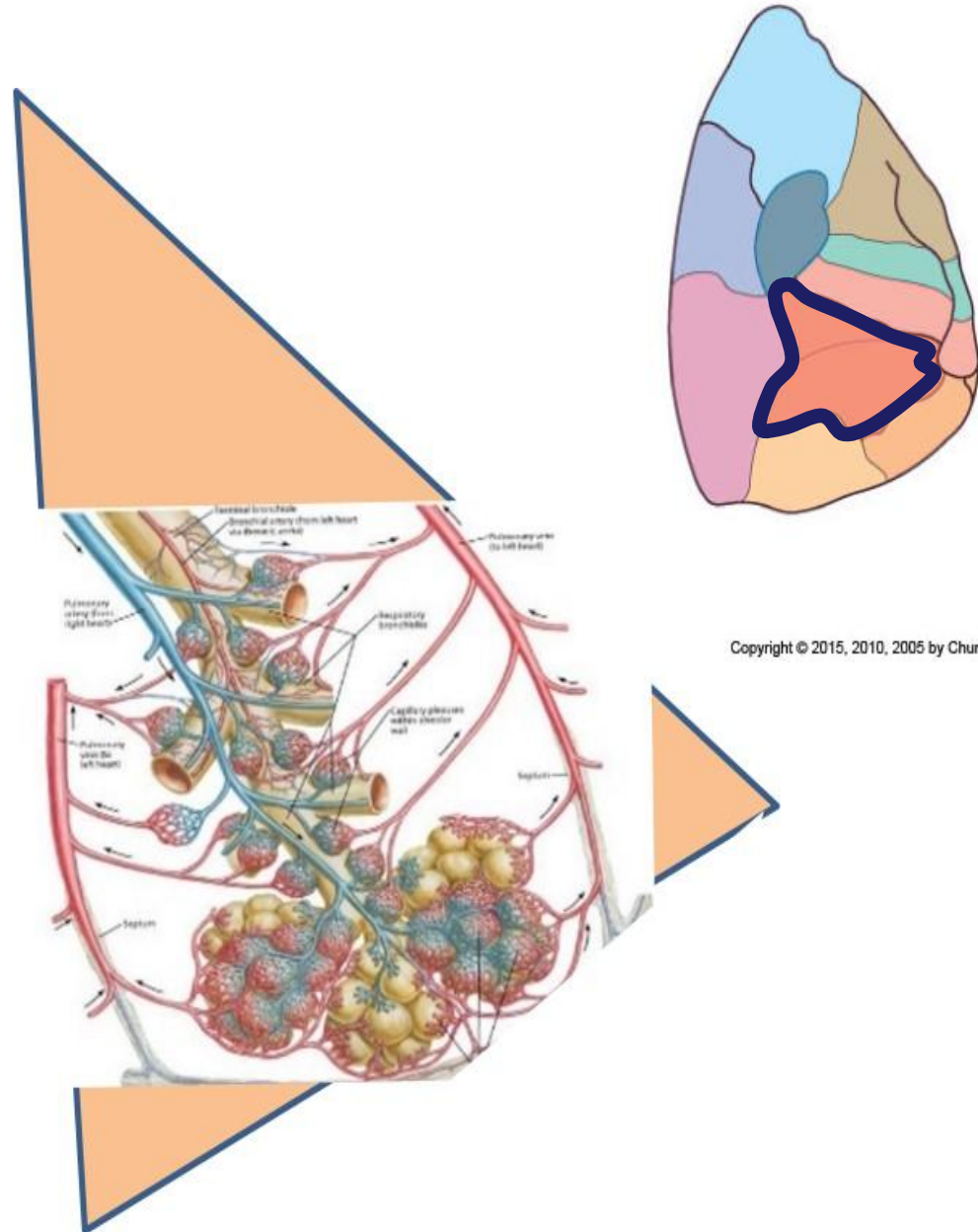
-Within each bronchopulmonary segment the segmental bronchi give rise to multiple smaller subdivisions to ultimately reach bronchioles.



- Aspirated/inhaled foreign objects tend to lodge in the wider, shorter and more vertical right bronchus and eventually in the right lower lobe.
- Each lobe segment has own arterial supply, segmental bronchus

Bronchopulmonary Segments

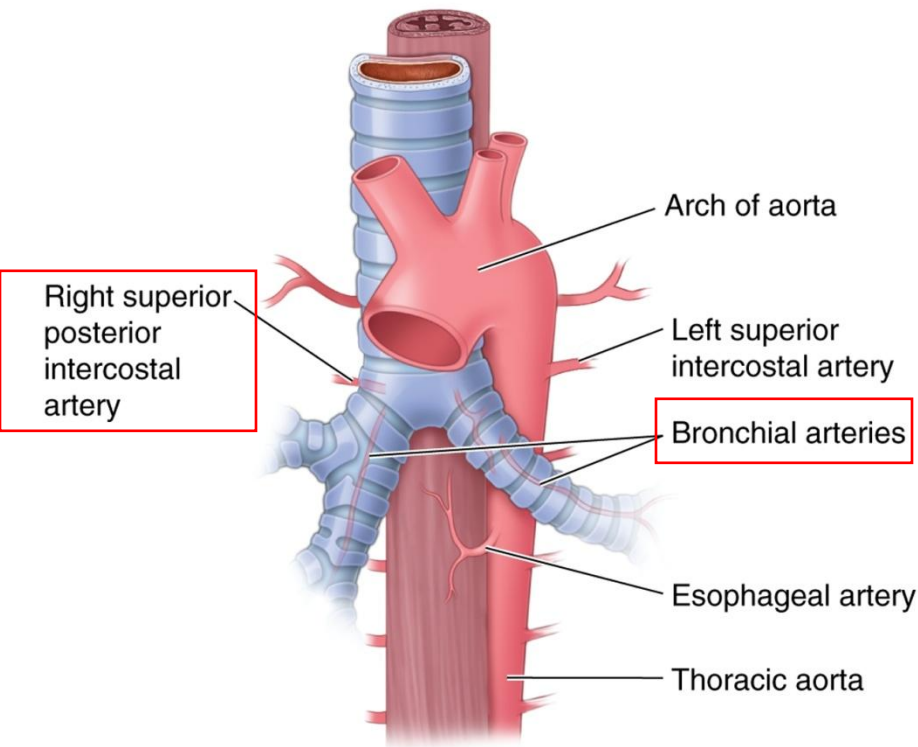
- Smallest functional unit of lung (approx. 10)
- Conical in shape
- Surgically resectable without affecting neighboring regions.
- Tertiary bronchi supply bronchopulmonary segments
- A branch of pulmonary artery accompanies the tertiary bronchi
- Drained by pulmonary veins



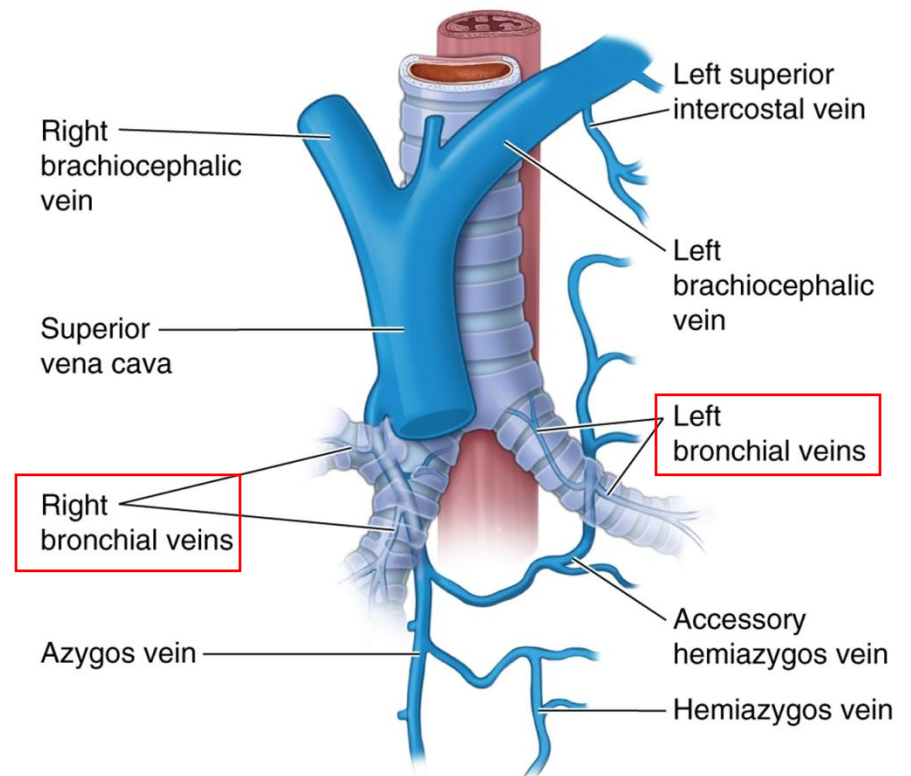
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LUNG- BLOOD CIRCULATION

- Bronchial vessels supply the bronchial tree and visceral pleura
 - Left bronchial arteries – direct branches of the aorta
 - Right bronchial artery from posterior intercostal artery
 - Bronchial veins drain into the hemiazygos (left bronchial) and azygos veins (right bronchial)

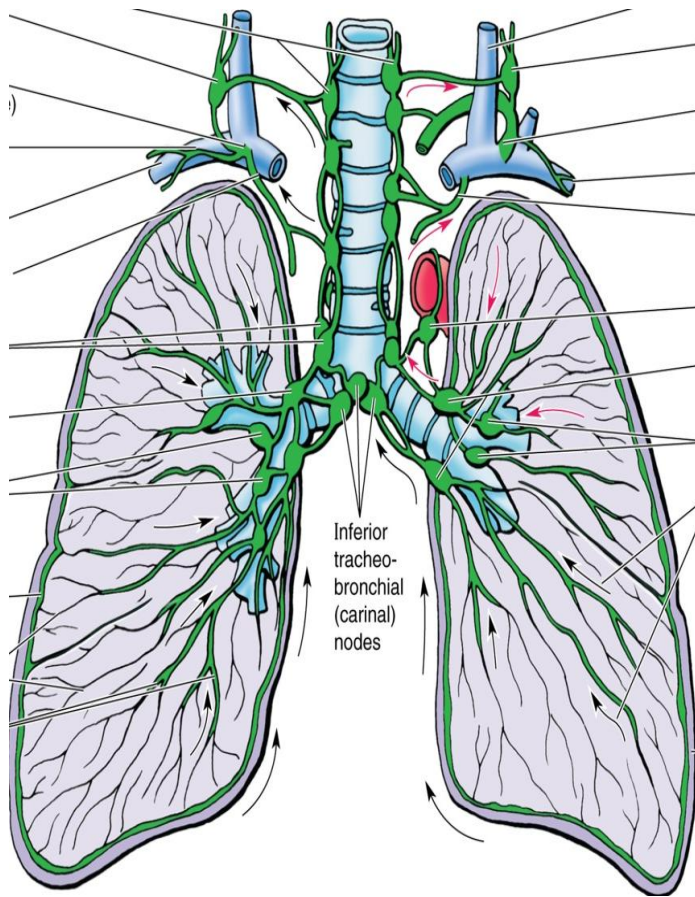


(A) Bronchial arteries



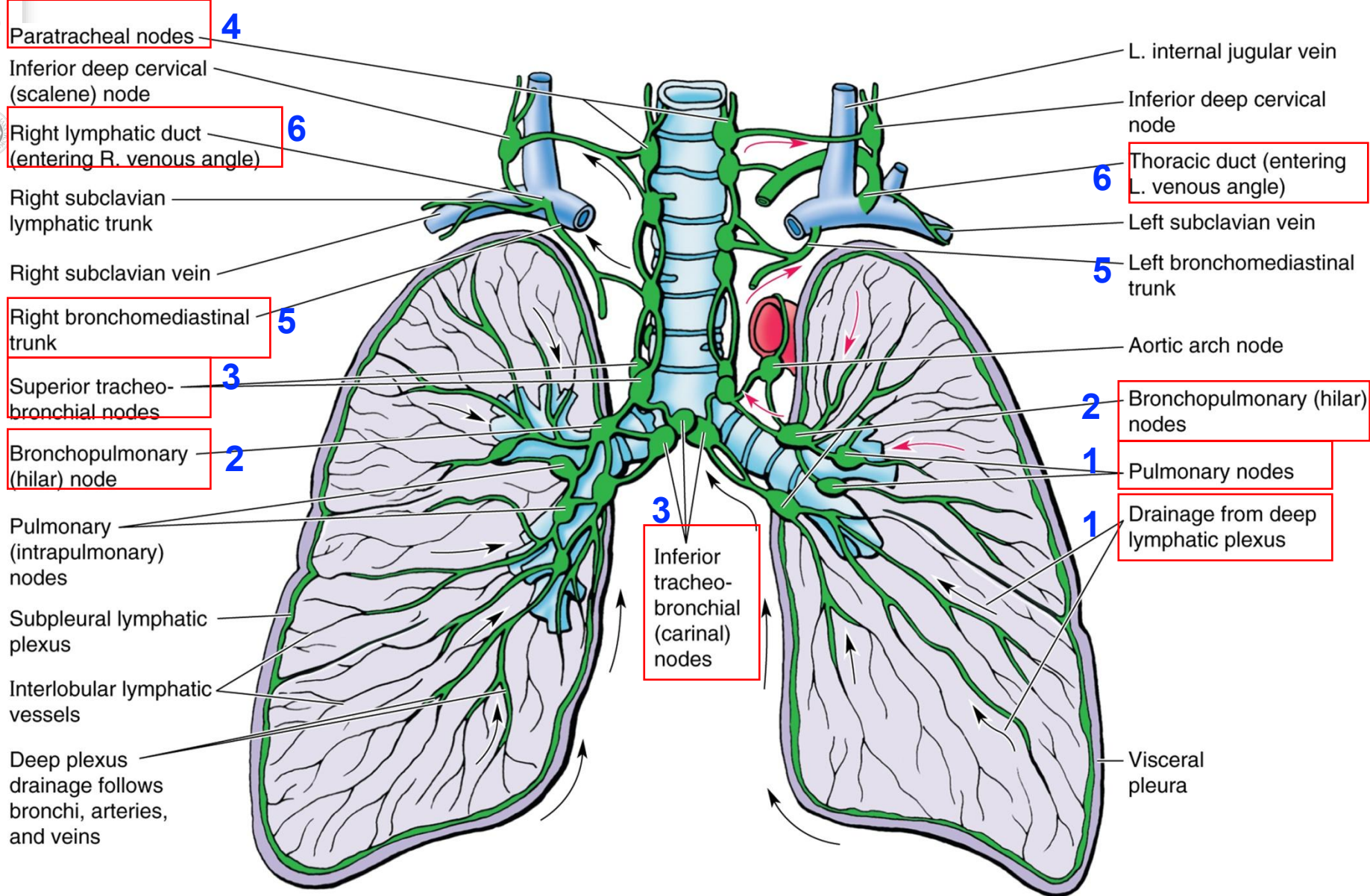
(B) Bronchial veins

Lymphatic Drainage



- 1 Intrapulmonary vessels & nodes
⇓
- 2 bronchopulmonary (hilar) nodes
⇓
- 3 tracheobronchial (carinal) nodes
⇓
- 4 paratracheal nodes
⇓
- 5 bronchomediastinal lymph trunk
⇓
- 6 right thoracic trunk/thoracic duct
⇓
- systemic venous system

Lymphatic Drainage



Innervation of the Lung

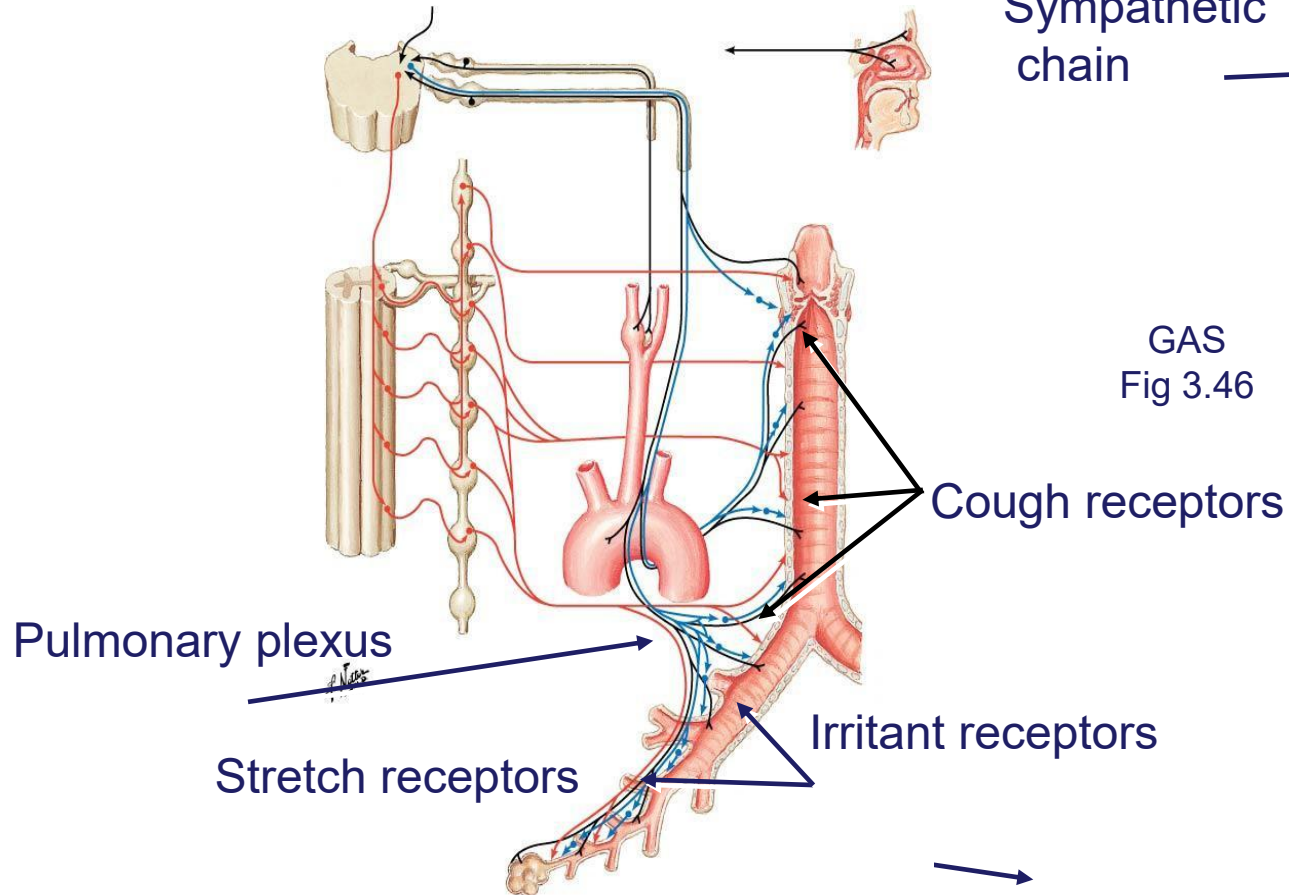
The lungs receive their innervation from the autonomic nervous system, comprising both sympathetic and parasympathetic components, which regulate various respiratory functions.

Parasympathetic Innervation: The vagus nerve (cranial nerve X) provides the parasympathetic innervation to the lungs. It stimulates bronchoconstriction, increases mucus secretion, and promotes vasodilation of pulmonary vessels. The vagus nerve forms the pulmonary plexus, which surrounds the bronchi and extends to the visceral pleura. [Kenhub+1Wikipedia+1](#)

Sympathetic Innervation: Sympathetic fibers originate from the upper thoracic segments of the spinal cord and synapse in the sympathetic ganglia. These fibers innervate bronchial blood vessels and submucosal glands, leading to bronchodilation and vasoconstriction. They also contribute to the pulmonary plexus, influencing airway tone and vascular tone. [Wikipedia+2Nature+2Oxford Research Encyclopedia+2](#)

- **Visceral Afferent Innervation:** Sensory fibers, primarily carried by the vagus nerve, transmit sensations such as stretch and irritation from the lungs to the central nervous system. These afferent fibers play a role in reflexes like coughing and bronchoconstriction in response to irritants.

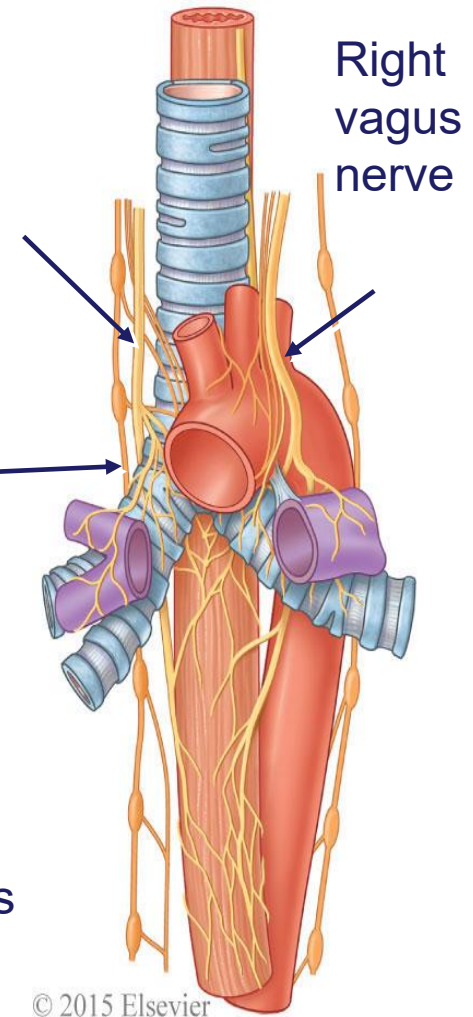
Innervation of the Lung



Left vagus nerve

Sympathetic chain

GAS
Fig 3.46



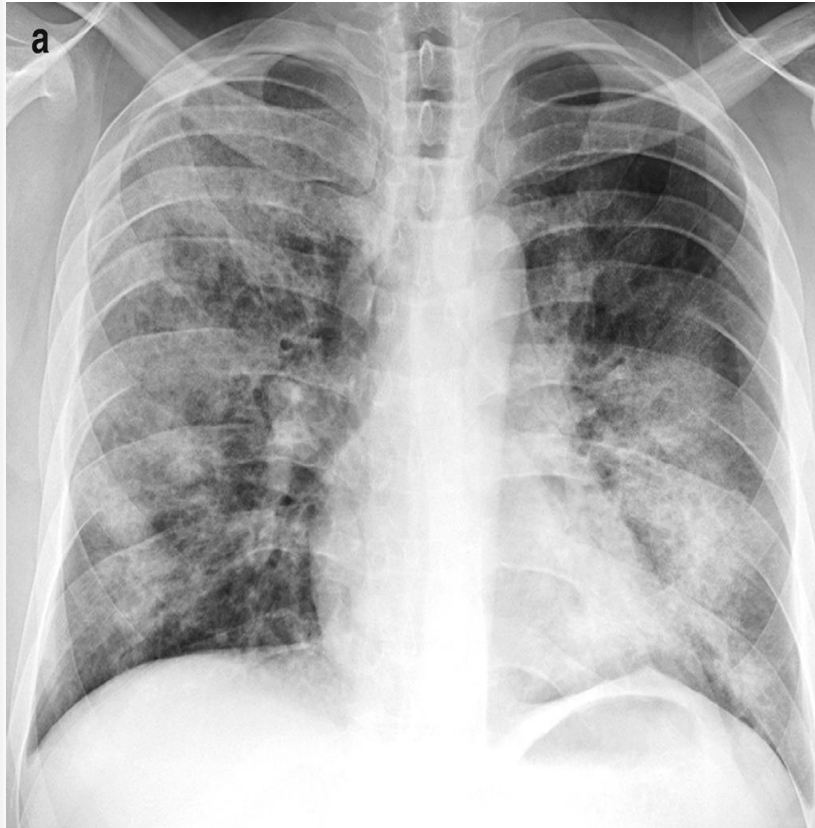
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PNEUMONIA

- Is an **inflammatory** condition of the **lung** primarily affecting the small air sacs known as **alveoli**.
- **Symptoms** typically include some combination of productive or dry cough, chest pain, fever, and difficulty breathing. The severity of the condition is variable.
- **Causes**
 - By infection with viruses or bacteria, and less commonly by other microorganisms.
 - Diagnosis is often based on **symptoms** and **physical examination**.
 - **Chest X-rays**, blood tests, and culture of the sputum may help confirm the diagnosis.
 - The disease may be classified by where it was acquired, such as community- or hospital-acquired
 - It can affect the whole lobe (lobar pneumonia) or only bronchioles and the surrounding alveoli (bronchopneumonia)

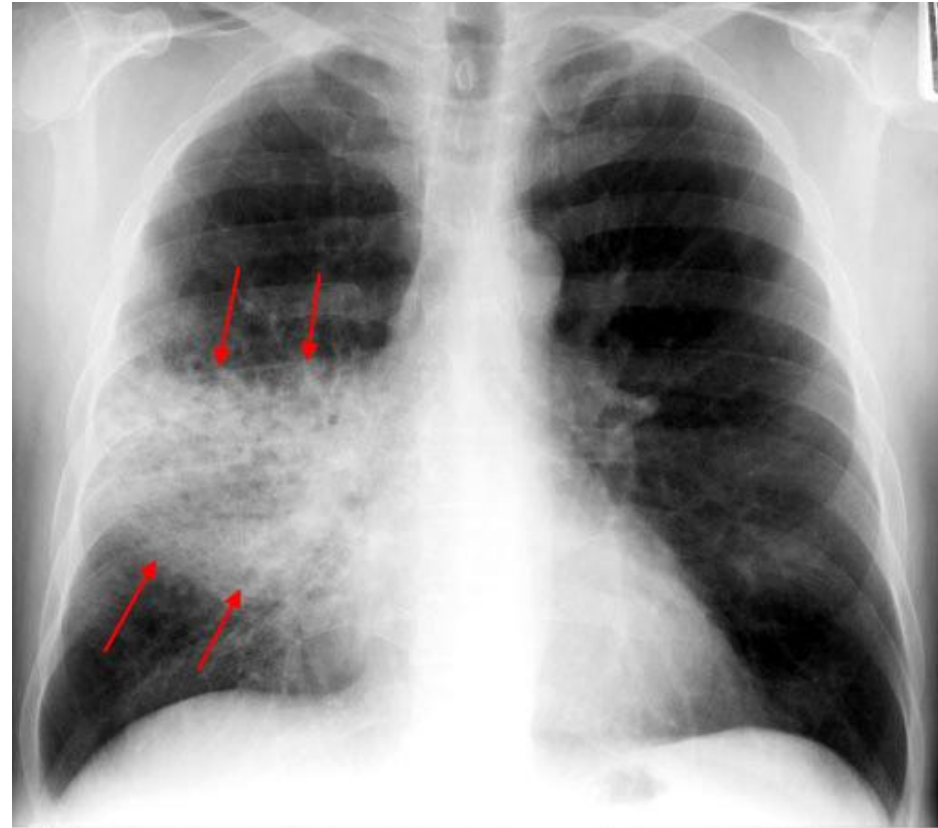
Bronchopneumonia



https://prod-images-static.radiopaedia.org/images/26900254/23e2fafbb64784be2579d79fb5dfab_big_gallery.jpeg

Lobar pneumonia

Patient with right middle lobe consolidation



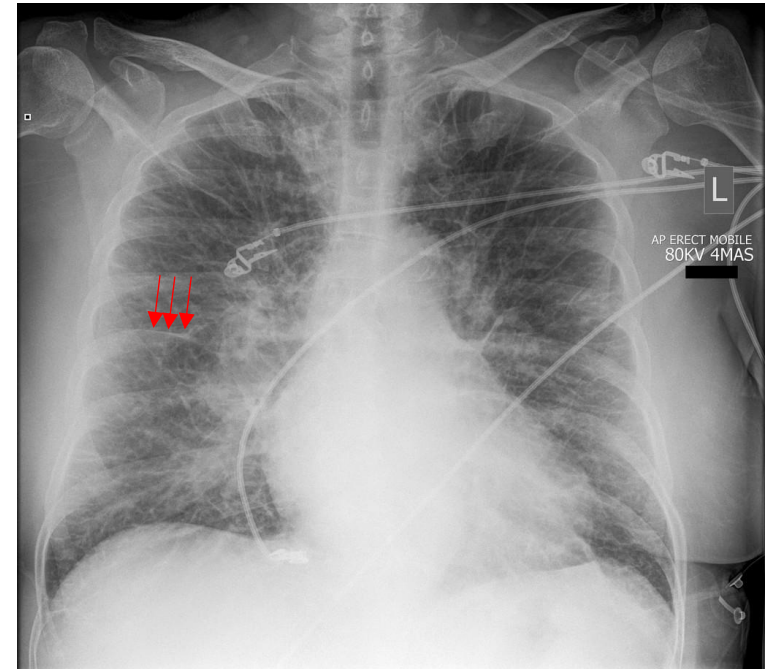
<http://medchrome.com/wp-content/uploads/2015/06/rml-pneumonia-pa1b.jpg>

ACUTE PULMONARY EDEMA

Acute Pulmonary Edema

- Collection of fluid in the alveoli.
- Usually due to cardiac failure, but can be due to cardiogenic shock, toxins, pneumonia or medications
- The most common symptom of pulmonary edema is difficulty breathing (dyspnea), but may include other symptoms such as coughing up blood (classically seen as pink or red, frothy sputum), excessive sweating, anxiety, and pale skin.

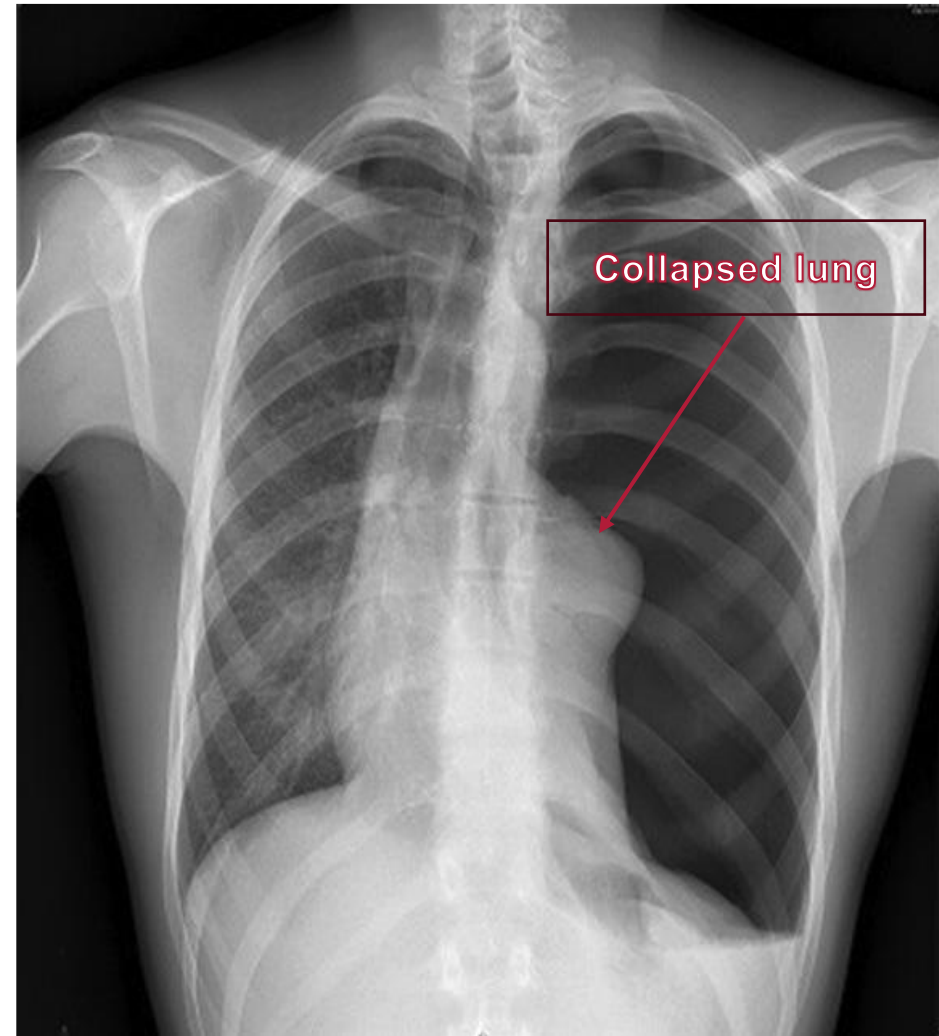
<https://radiopaedia.org/cases/acute-pulmonary-oedema-5?lang=us>



Hazy perihilar opacity, right > left. Kerley lines are present. These represent thickening of interlobular (segmental) septum which become more prominent with edema of the connective tissue of the lung.

Atelectasis

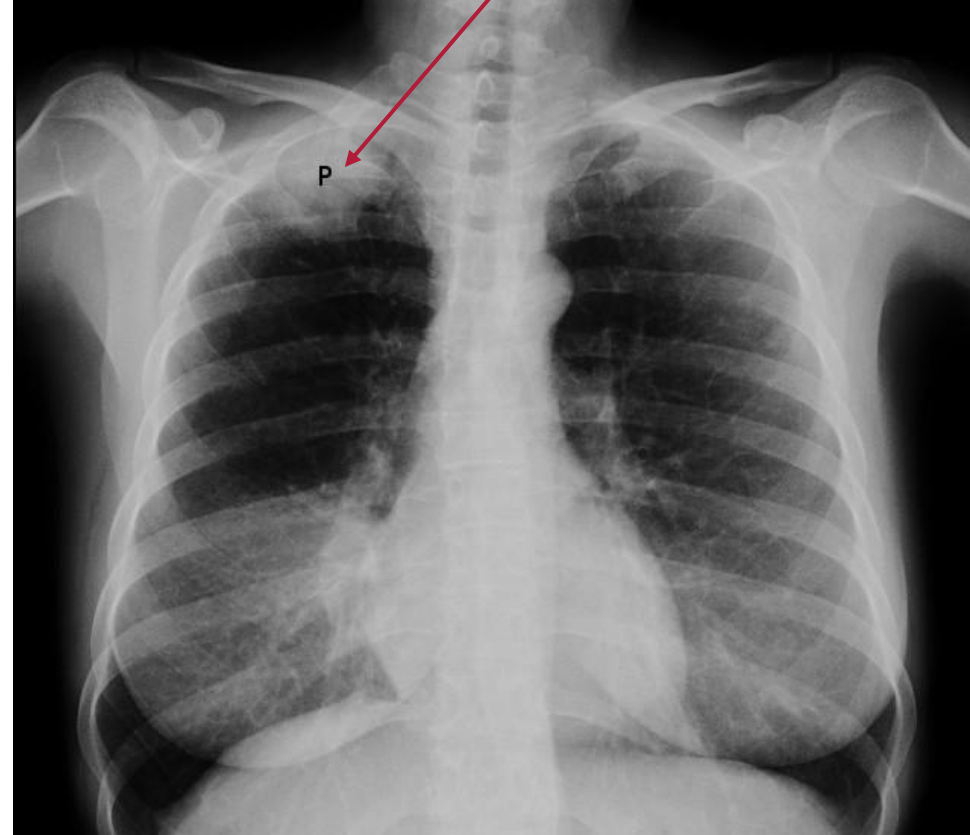
- **Atelectasis** is the collapse or closure of a lung resulting in reduced or absent gas exchange. It is usually unilateral, affecting part or all of one lung.
- The **most common cause** is post-surgical atelectasis, characterized by splinting, i.e. restricted breathing after abdominal surgery. Atelectasis develops in 75–90% of people undergoing general anesthesia for a surgical procedure



Pancoast tumor

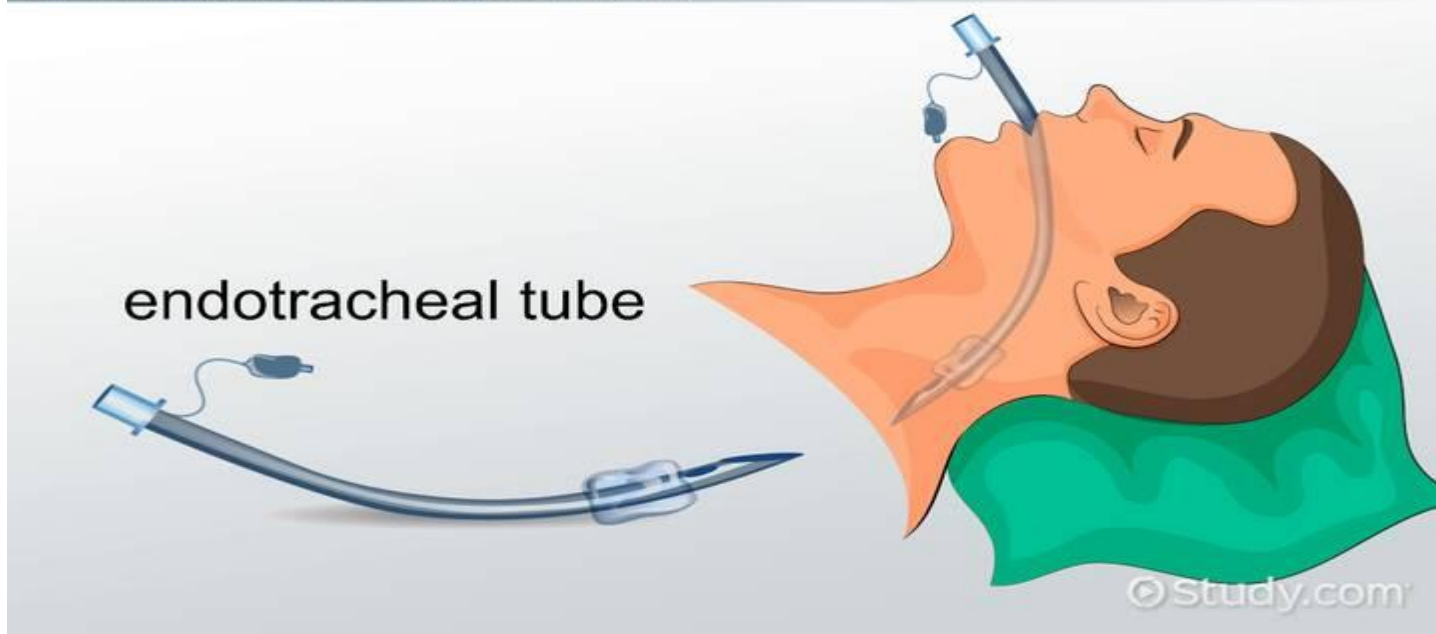
- A **Pancoast tumor** is a tumor of the apex of the lung. It is a type of lung cancer defined primarily by its location situated at the top end of either the right or left lung. It typically spreads to nearby tissues such as the ribs and vertebrae.
- Most Pancoast tumors are non-small-cell lung cancers.

Lung mass

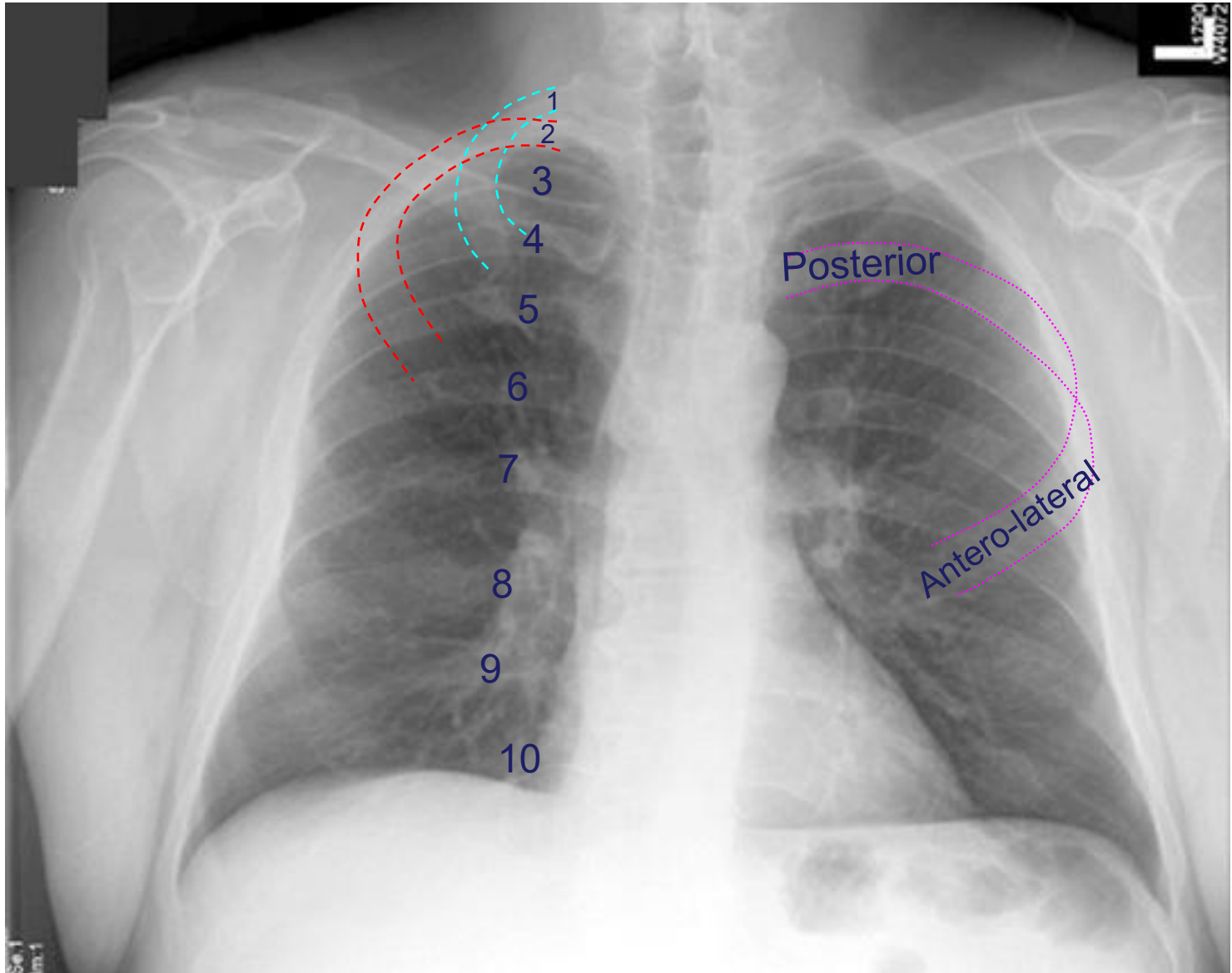


- The insertion of an endotracheal tube, also known as intubation, is a medical procedure that is performed to secure a patient's airway
- The endotracheal tube is inserted through the mouth or nose, passes through the pharynx and larynx, and enters the trachea.
- **Clinical uses:**
 - Provision of oxygen
 - Provision of drugs (anesthetic agents)

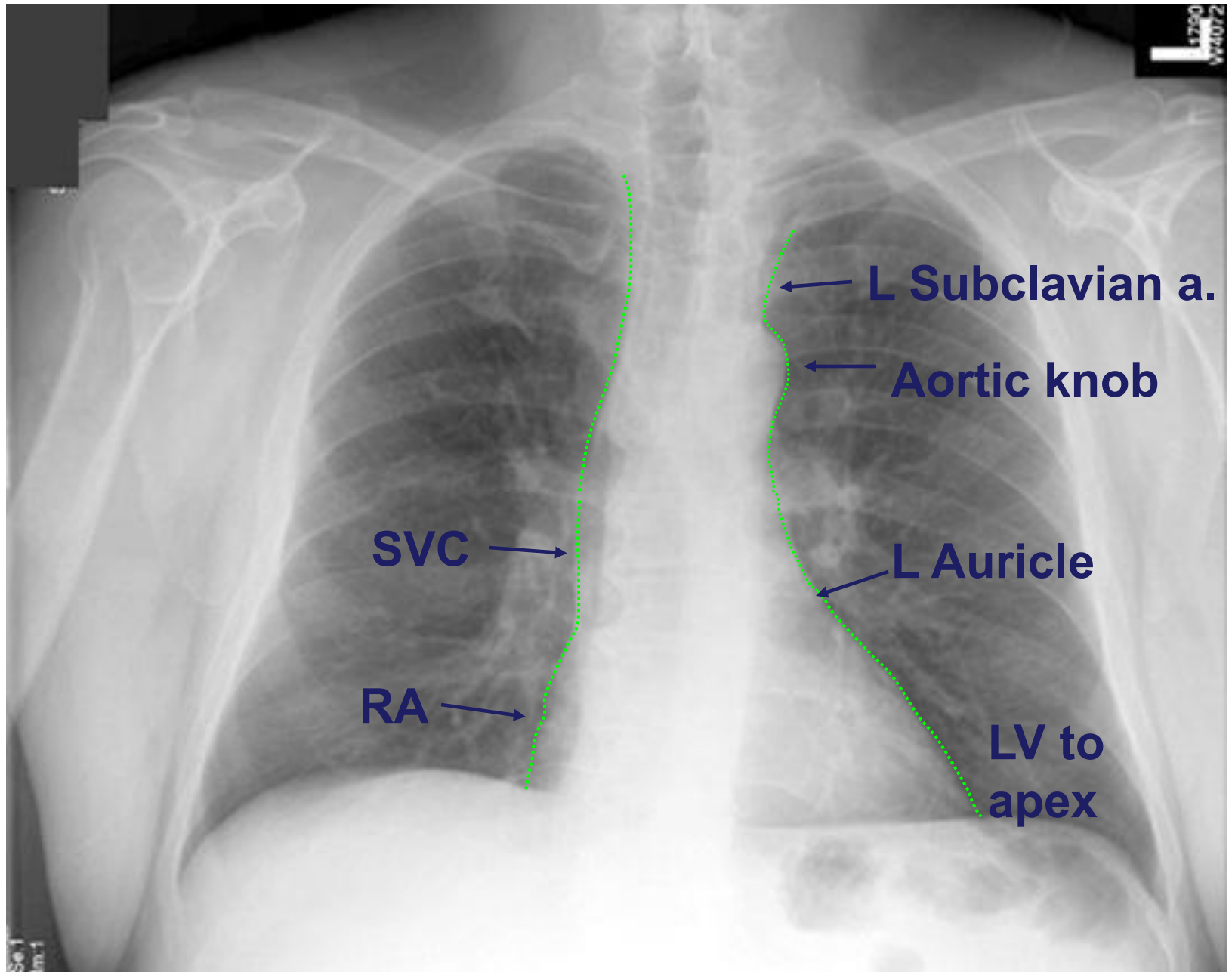
ENDOTRACHEAL TUBE INDICATIONS



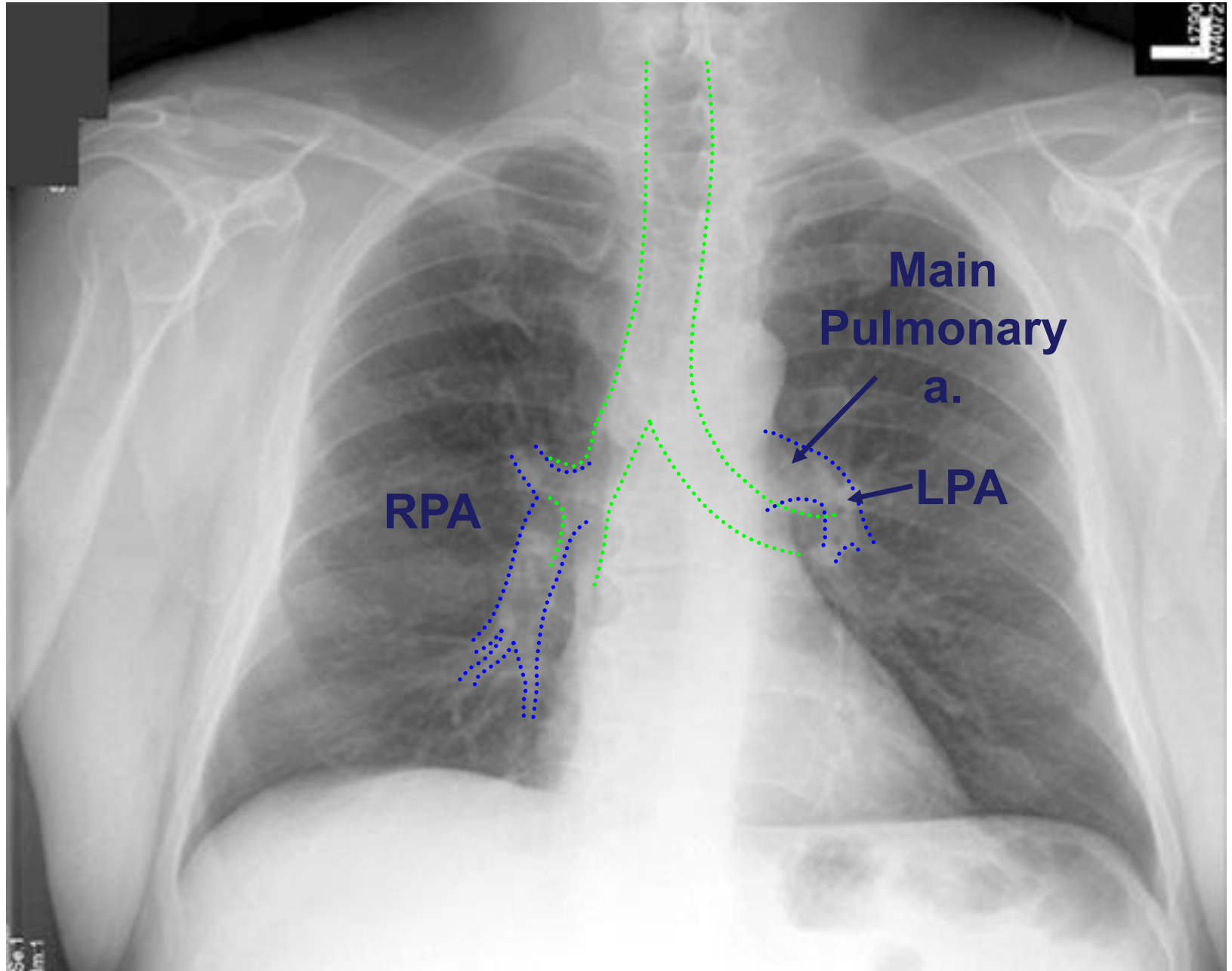
Inspiration – 9-10 ribs should be visible



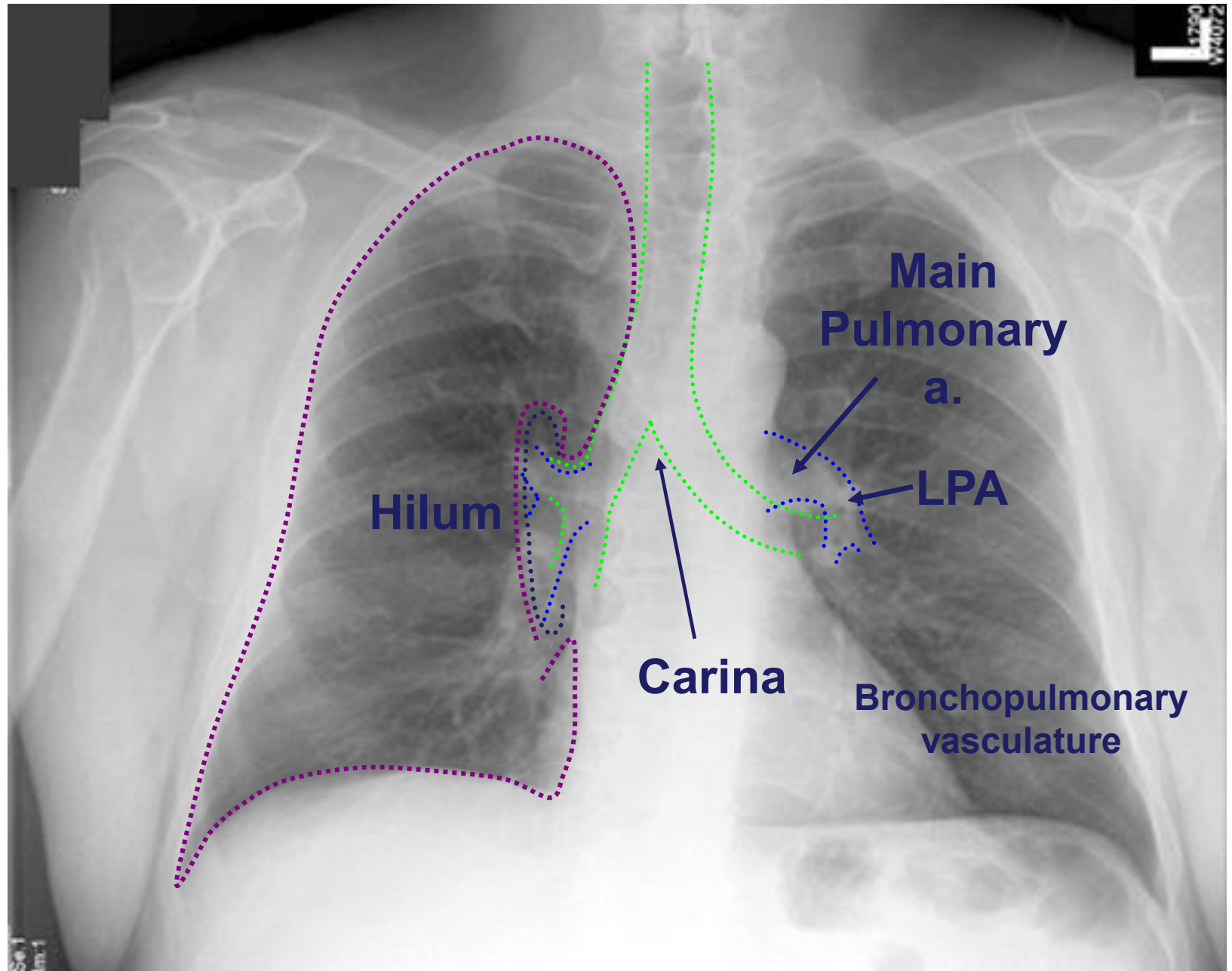
Mediastinal Silhouette



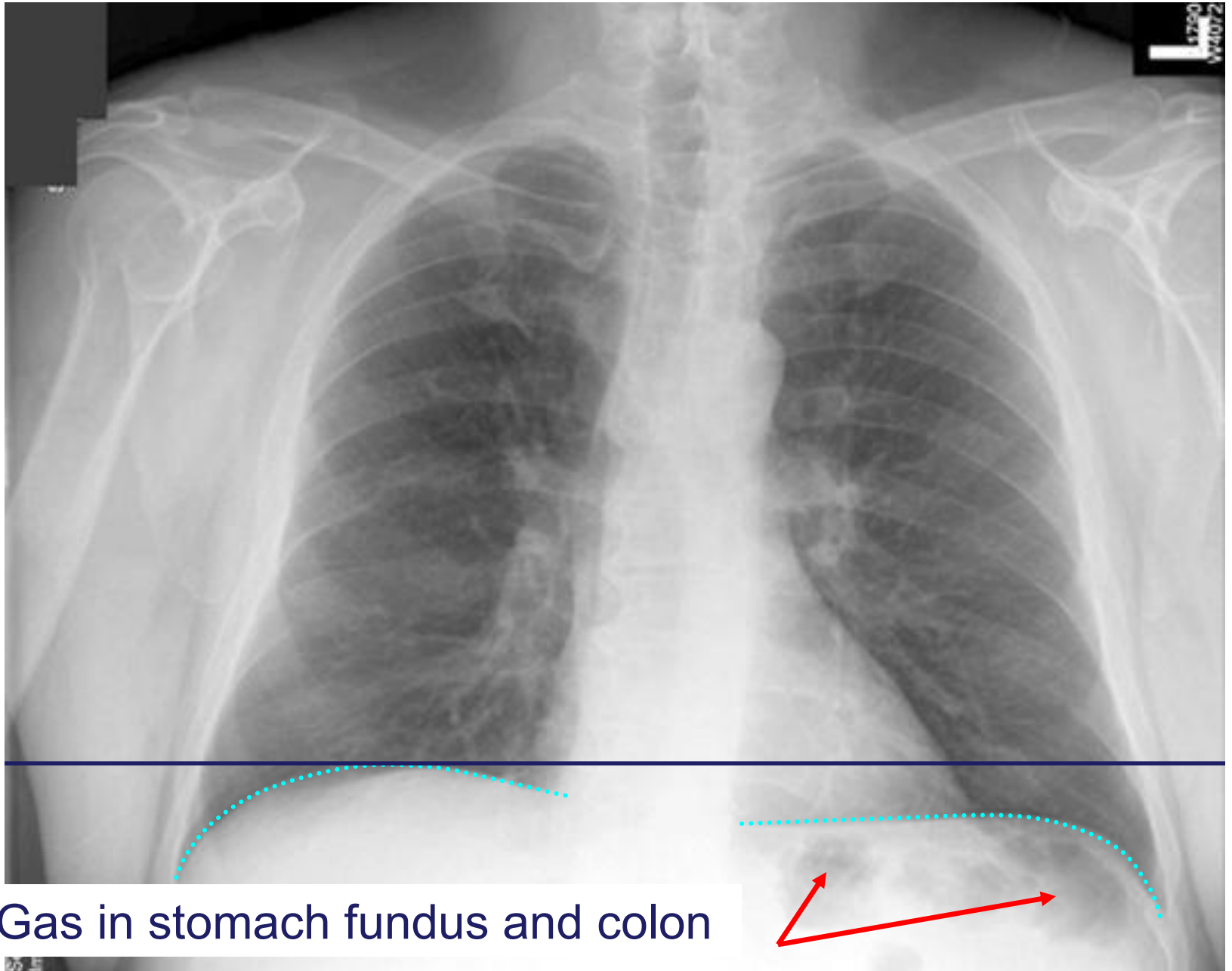
Airways and Hila



Airways and Hila



Hemi-diaphragms and upper abdomen



Thank you

Questions?

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